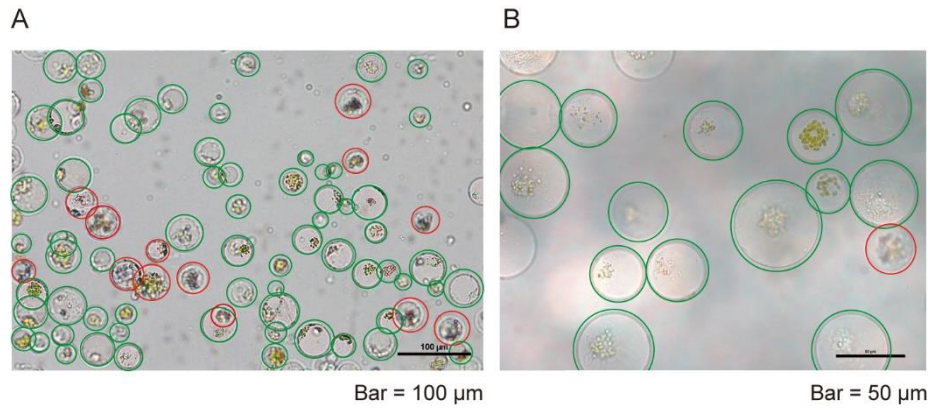
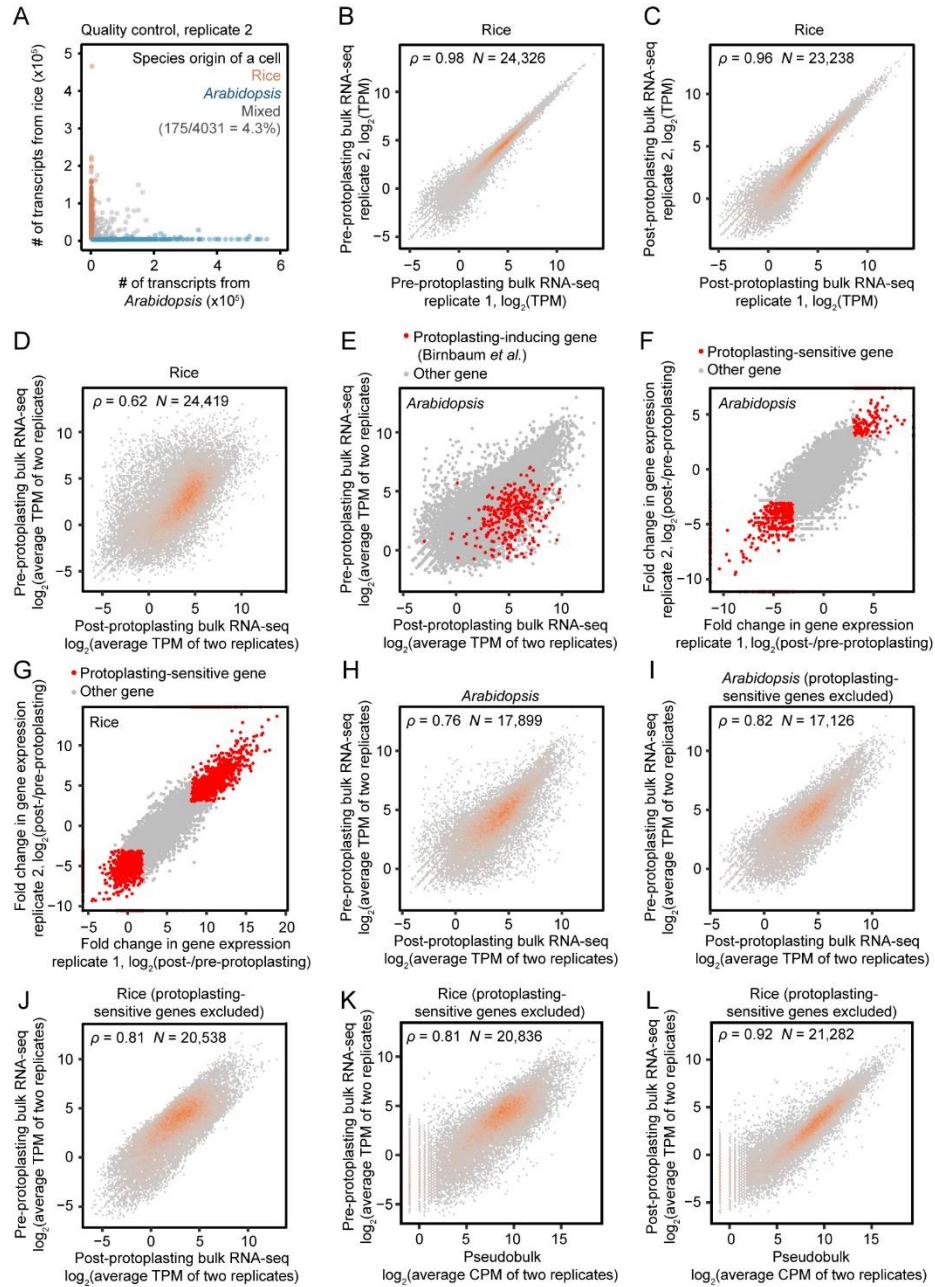




**Fig. S1. The quality of isolated protoplasts.**

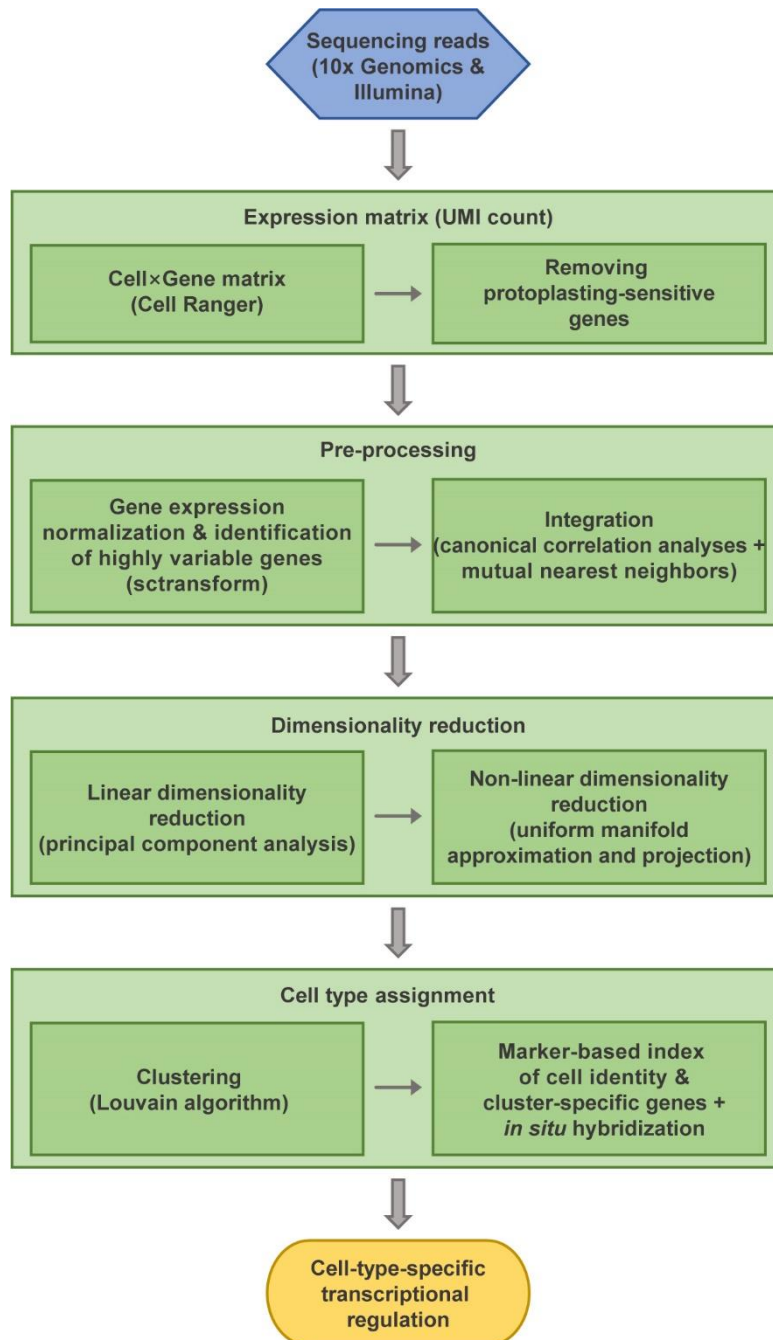
**A:** The viability of protoplasts was detected using 0.4% trypan blue staining. Dead protoplasts were stained in blue and labeled by red circles. Viable protoplasts were labeled by green circles. Scale bar represents 100  $\mu\text{m}$ . **B:** Zoomed-in view is shown. Scale bar represents 50  $\mu\text{m}$ .



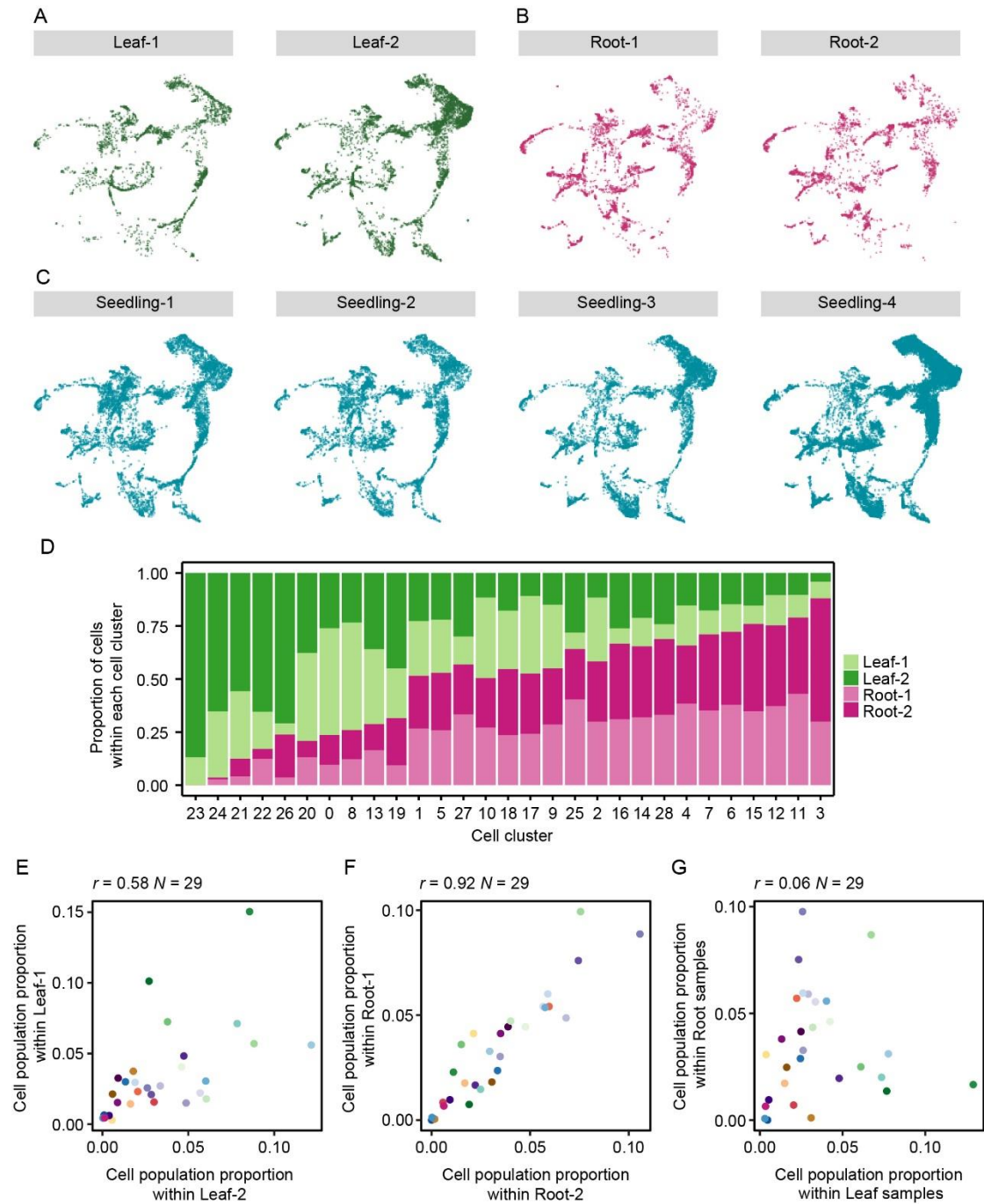
**Fig. S2. Additional plots for scRNA-seq quality control.**

**A:** A scatter plot shows the number of *Arabidopsis* transcripts versus rice transcripts for each cell. Blue and salmon dots represent cells that were determined as *Arabidopsis* and rice cells, respectively. Approximately 4.3% of cells were determined as “mixed” cells, which are indicated by grey dots. The result of replicate 2 is shown; that of the other biological replicate is shown in Fig. 1C. **B:** The density plot shows the rice gene expression levels of two replicates for the pre-prototyping bulk RNA-seq ( $P < 2.2 \times 10^{-16}$ , Spearman’s correlation). Red (or grey) dots represent a relatively

high (or low) density of genes. Gene expression levels are in the unit of transcript per million (TPM). **C**: Similar to (B), for post-protoplasting bulk RNA-seq ( $P < 2.2 \times 10^{-16}$ , Spearman's correlation). **D**: The density plot shows the rice gene expression levels (average of two replicates) between pre- and post-protoplasting bulk RNA-seq data ( $P < 2.2 \times 10^{-16}$ , Spearman's correlation). Red (or grey) dots represent a relatively high (or low) density of genes. **E**: The expression levels (the average of two replicates) of 342 reported protoplasting-inducing genes in *Arabidopsis* (by Birnbaum *et al.*, showing in red) in our pre- and post-protoplasting bulk RNA-seq data. **F-G**: Scatter plots show the protoplasting-sensitive genes in *Arabidopsis* (F) and rice (G) that identified in this study (showing in red). **H**: The density plot shows the *Arabidopsis* gene expression levels (the average of two replicates) between pre- and post-protoplasting bulk RNA-seq samples, before excluding protoplasting-sensitive genes ( $P < 2.2 \times 10^{-16}$ , Spearman's correlation). Red (or grey) dots represent a relatively high (or low) density of genes. **I-J**: The density plots show *Arabidopsis* (I) or rice (J) gene expression levels (the average of two replicates) between pre- and post-protoplasting bulk RNA-seq samples, after excluding protoplasting-sensitive genes ( $P < 2.2 \times 10^{-16}$  for both, Spearman's correlation). **K-L**: The density plots show rice gene expression levels (the average of two replicates) between pseudobulk and pre-(K) or post-protoplasting (L) samples, after excluding protoplasting-sensitive genes ( $P < 2.2 \times 10^{-16}$  for both, Spearman's correlation).



**Fig. S3.** A schematic shows the computational workflow of scRNA-seq data normalization, integration, dimensionality reduction, clustering, and cell-type assignment.



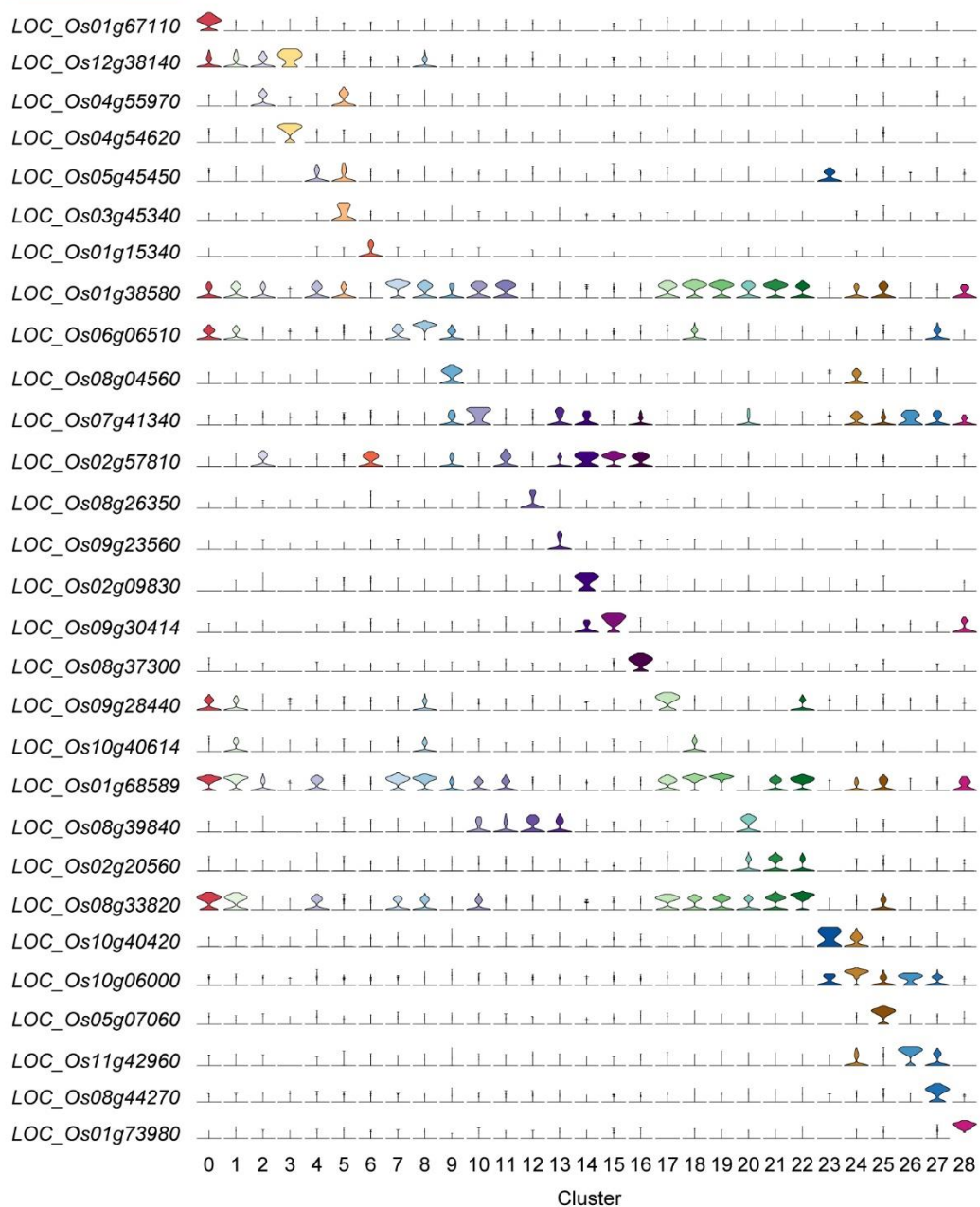
**Fig. S4. Cell population features of Leaf and Root samples.**

**A–C:** Visualization of Leaf, Root, and Seedling samples on the same UMAP space. Dots represent individual cells, colored according to the materials. **D:** The proportion of cells within each cell cluster for Leaf and Root samples. **E:** The scatter plot shows the proportion of cells in a sample that belongs to a cell cluster, between the two replicates of Leaf samples. Each dot represents one of the 29 cell clusters, colored

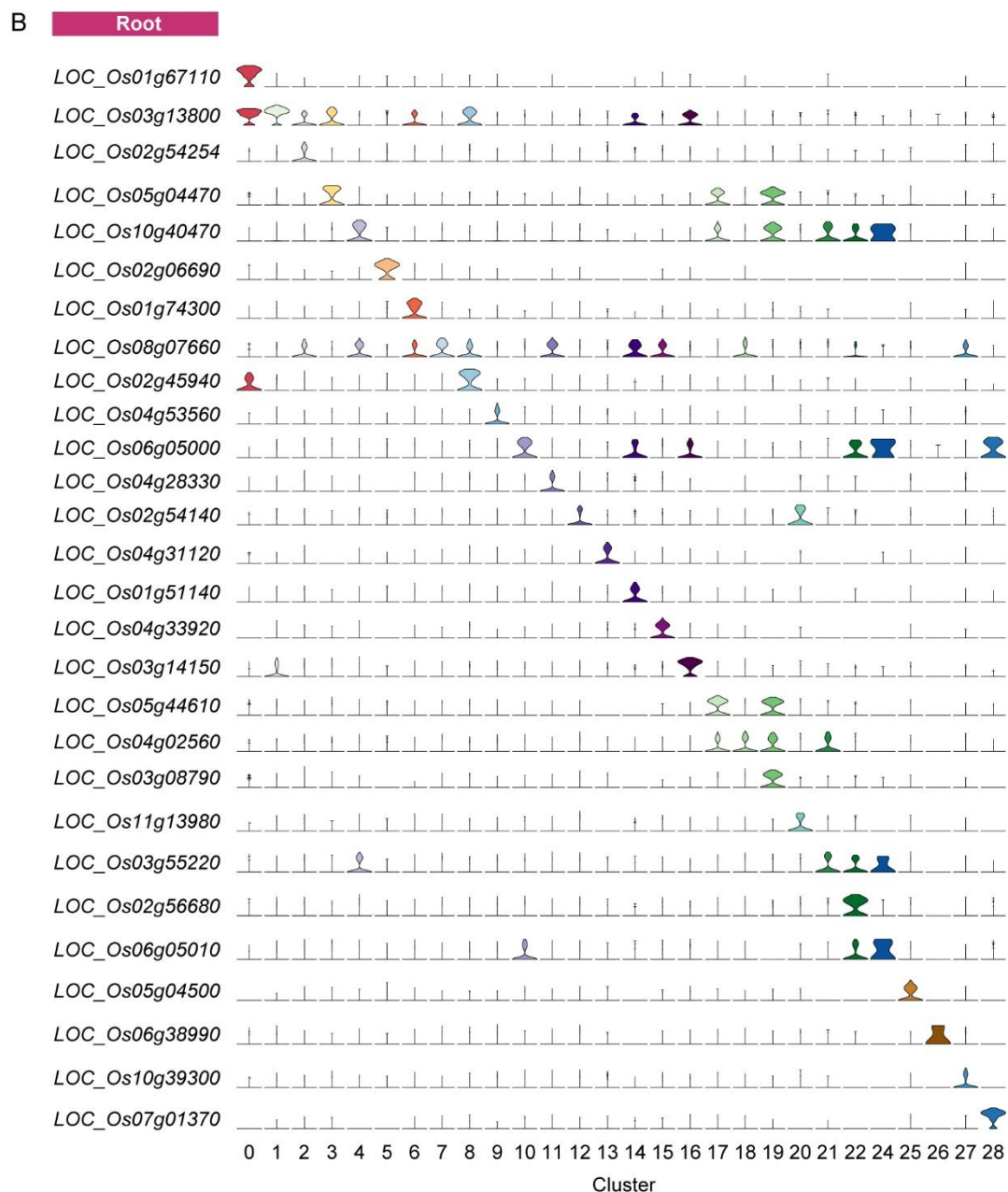
according to Fig. 2B. The Pearson's correlation coefficient  $r$  is shown on the top ( $P < 1 \times 10^{-3}$ ). **F**: Similar to (E), for Root samples ( $P < 1 \times 10^{-12}$ , Pearson's correlation). **G**: Similar to (E), between Root and Leaf samples (cells of two replicates combined;  $P = 0.75$ , Pearson's correlation).

A

Leaf

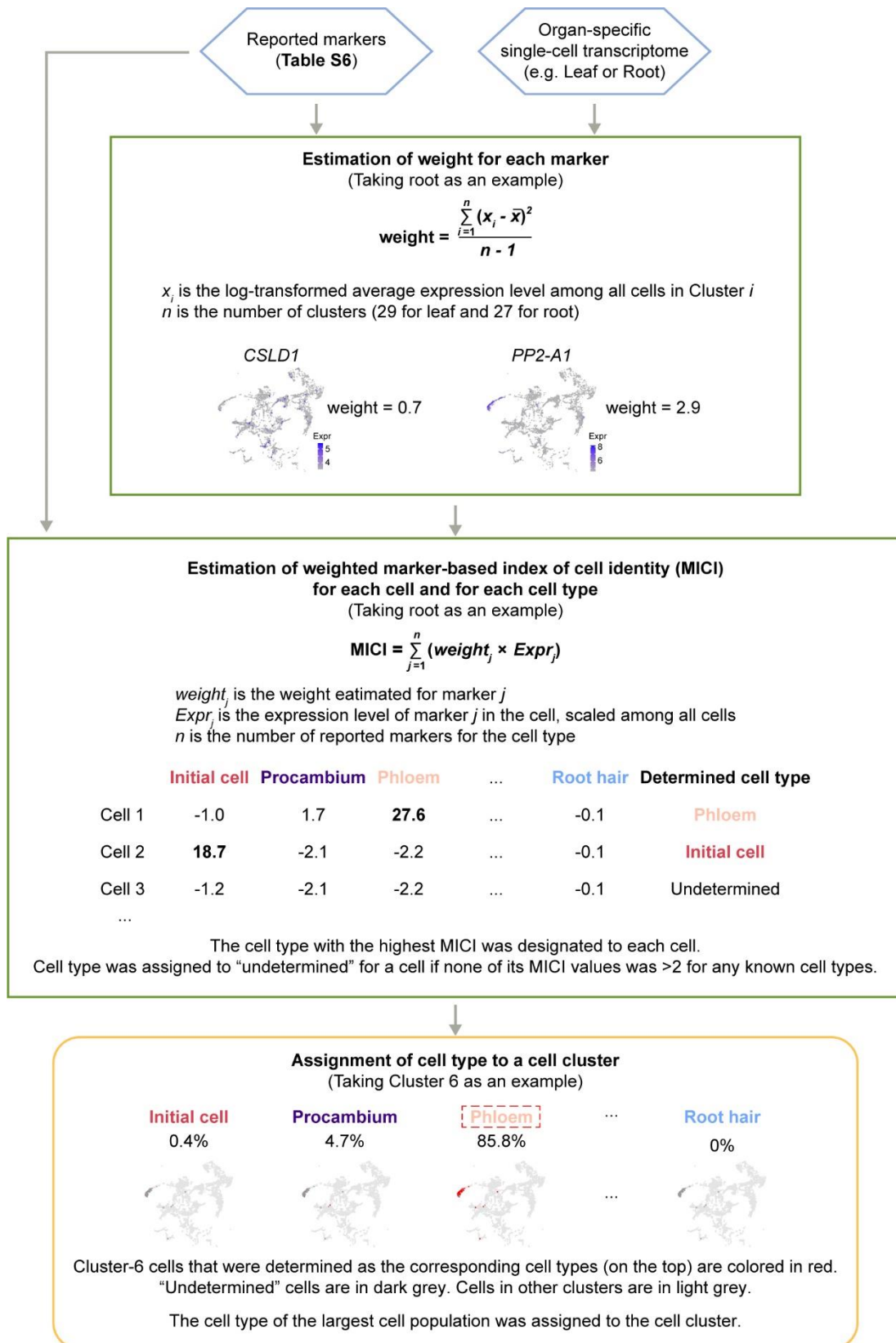




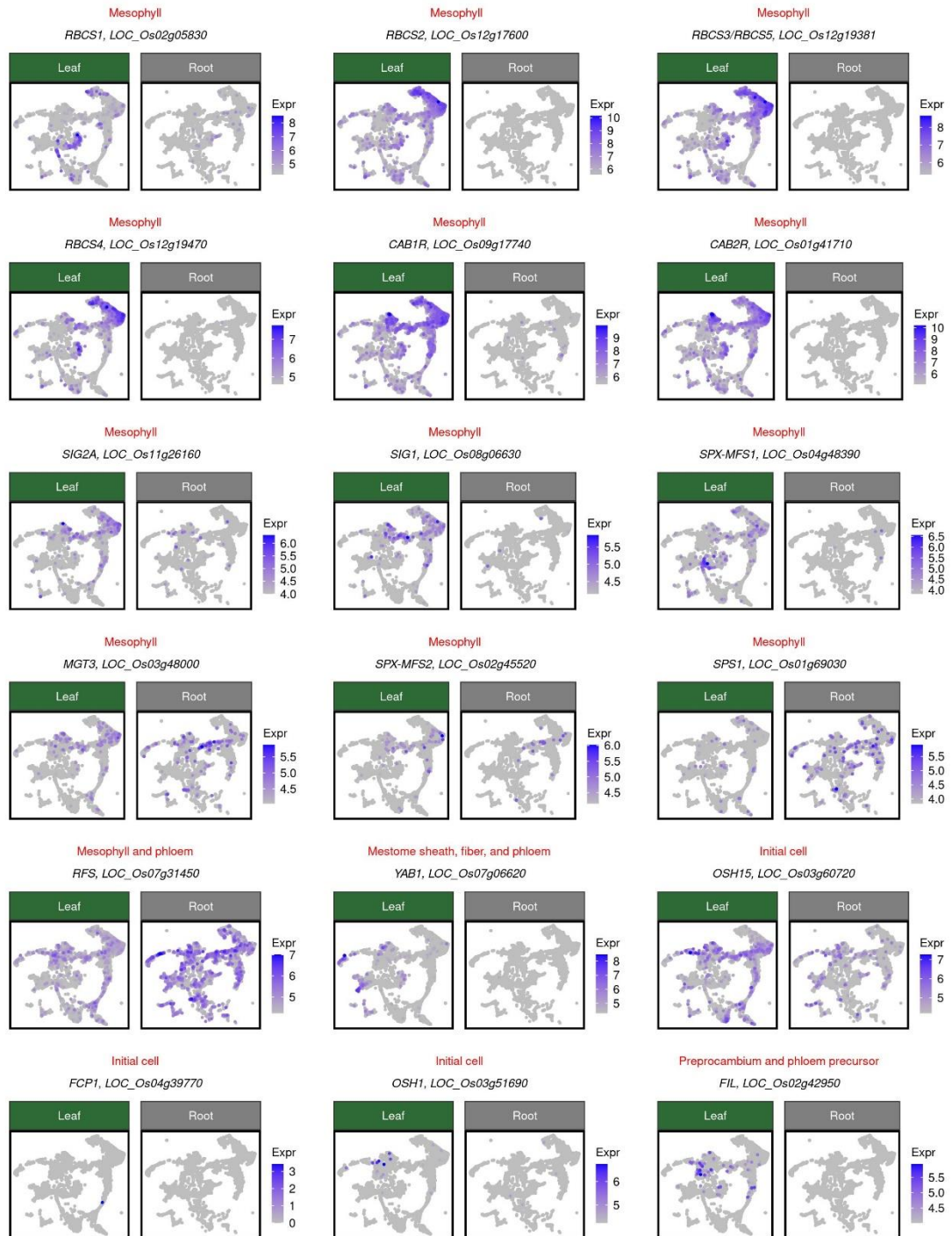


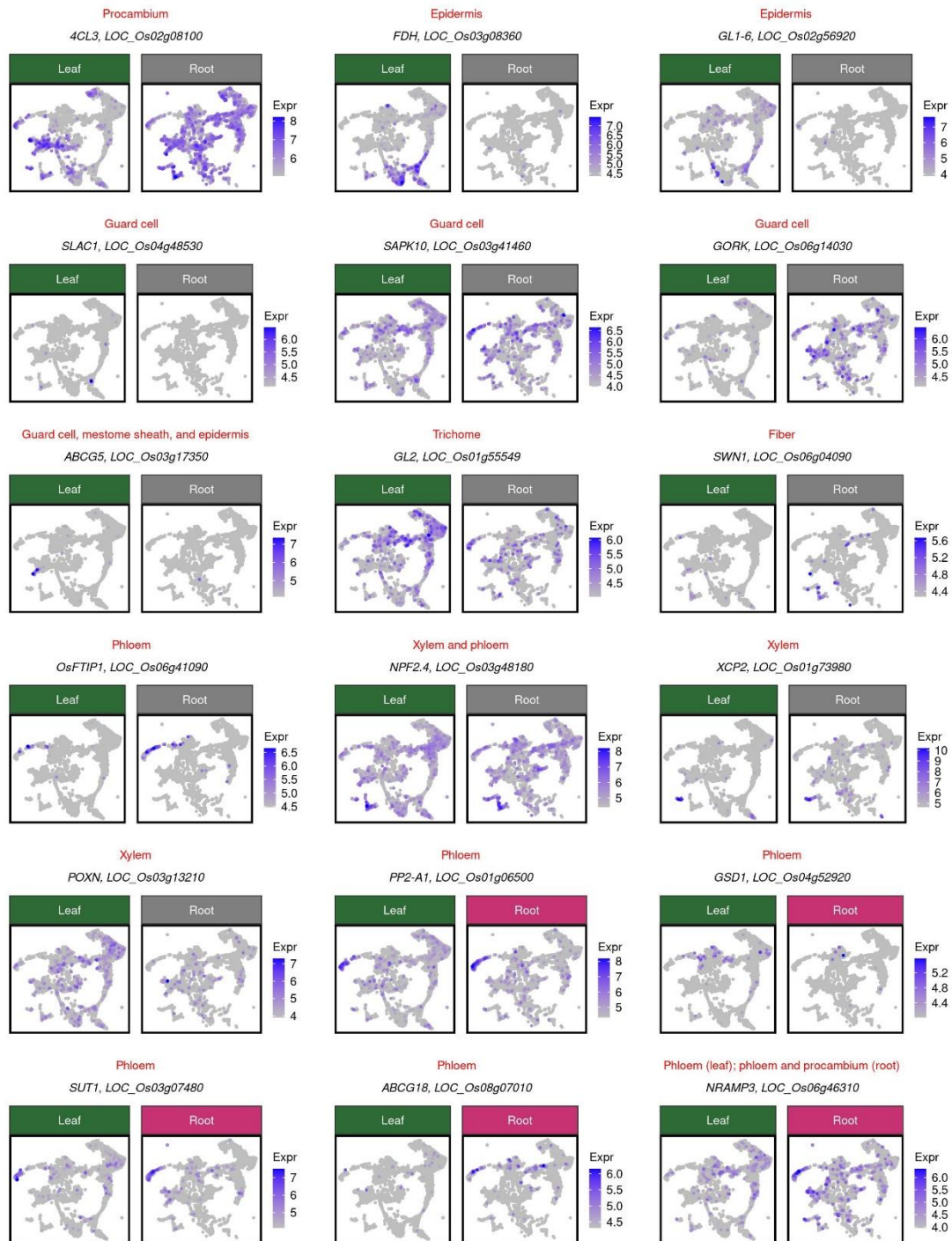
**Fig. S5. The violin plots showing expression specificity for cluster-specific genes (cells of the two replicates combined for Leaf or Root samples). There were no cells in Cluster 23 in the Root samples, which is therefore omitted in (B).**

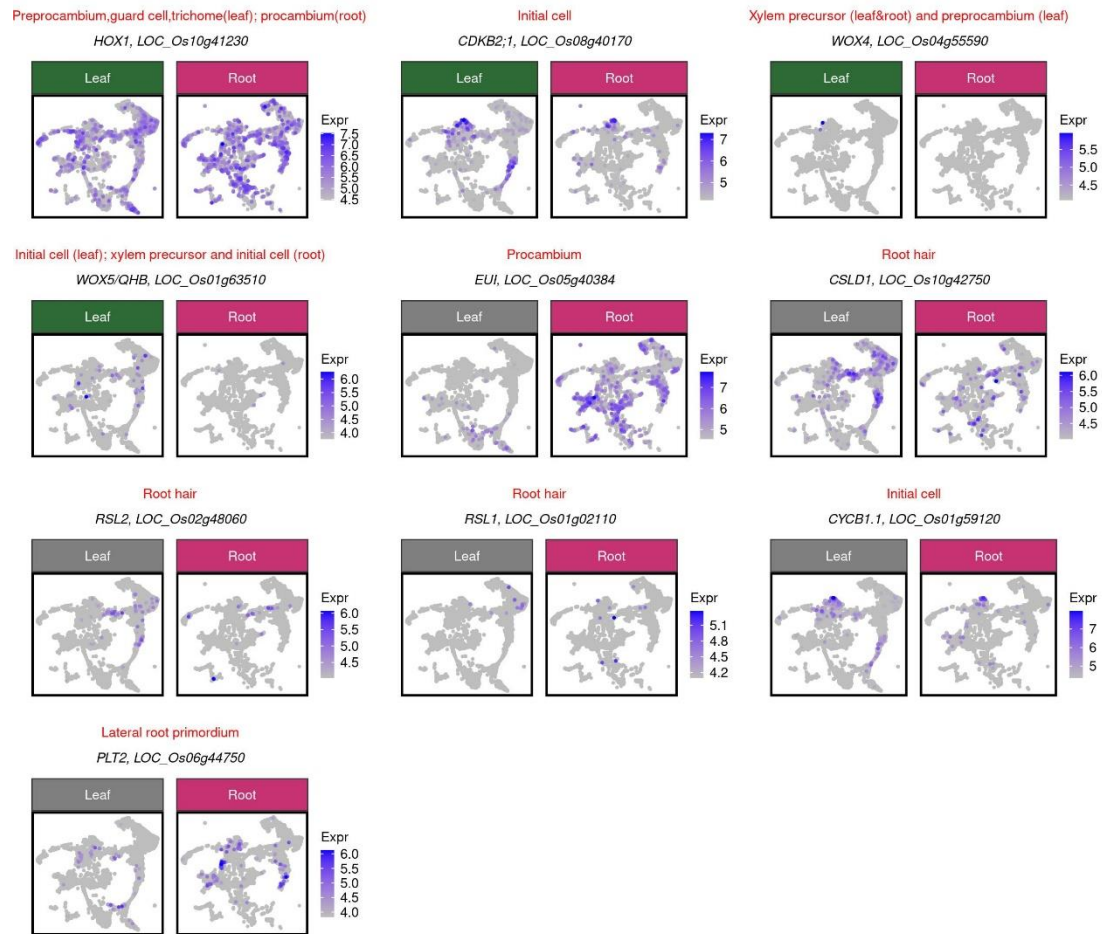




**Fig. S6. A schematic shows the computational workflow of the assignment of cell types to clusters by MICI.**

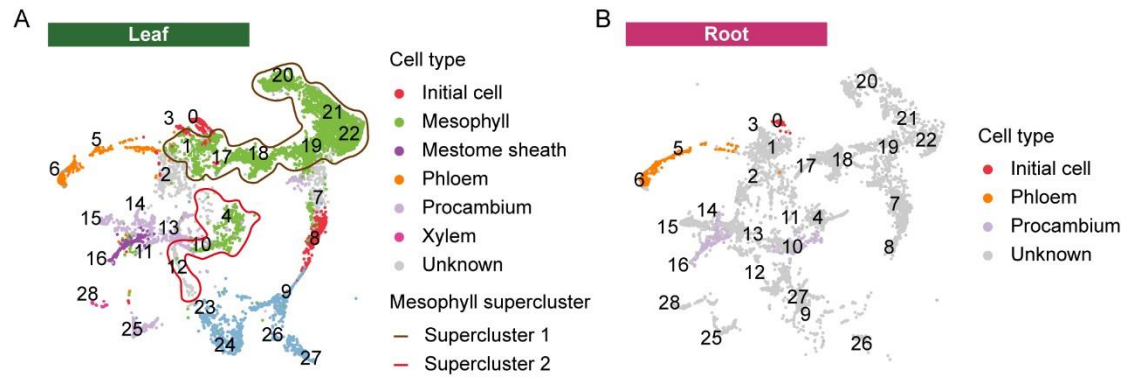






**Fig. S7. UMAP visualization of 46 cell-type marker genes previously reported.**

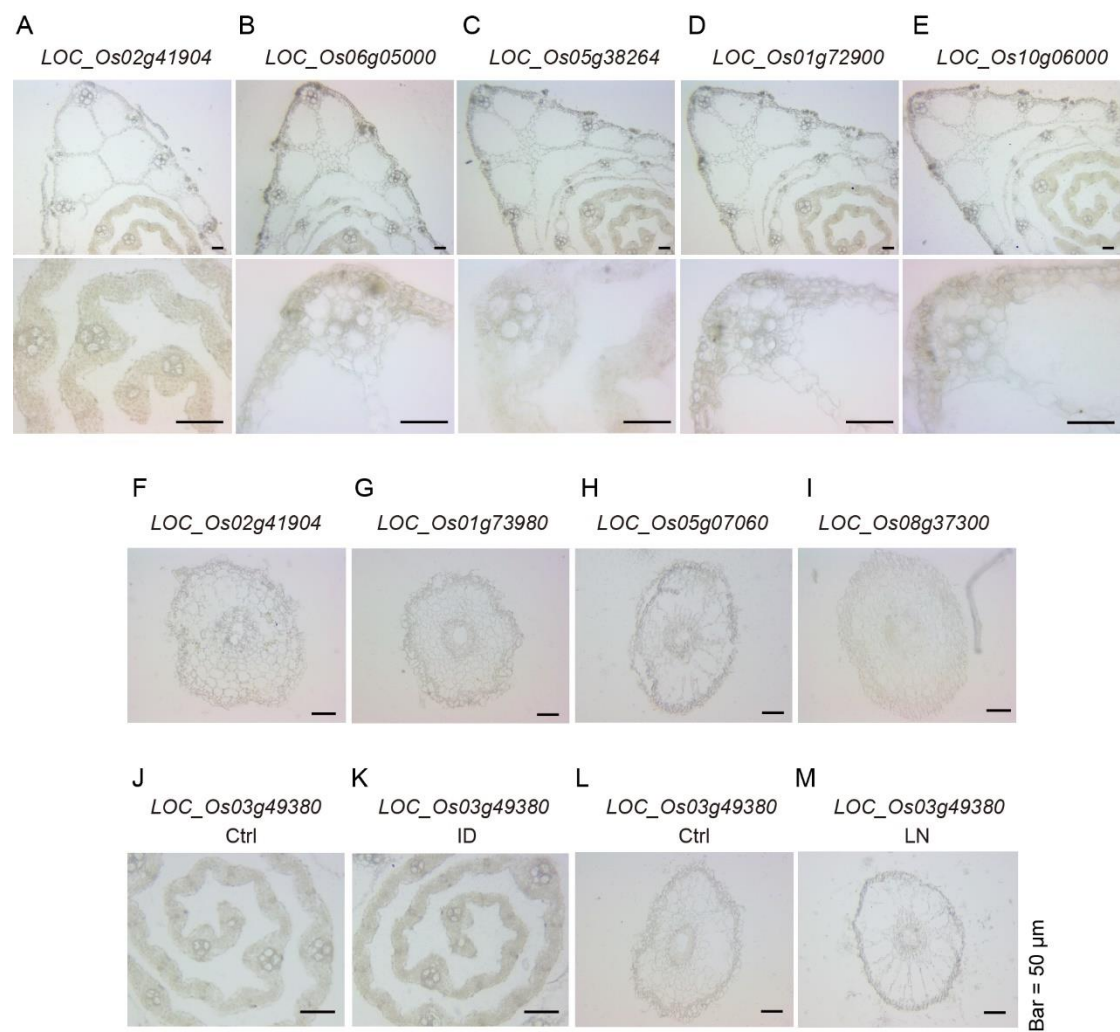
Dots represent individual cells, colored according to the normalized gene expression level,  $\log(\text{CPM} + 1)$ . For each gene, the reported cell types, gene symbols, and gene IDs are shown. The 40 and 15 marker genes for the leaf and the root (nine being shared), respectively, are highlighted in the subtitle. Detailed information on these 46 marker genes is provided in Table S6.



**Fig. S8. The initial assignment of cell types to clusters by MICI.**

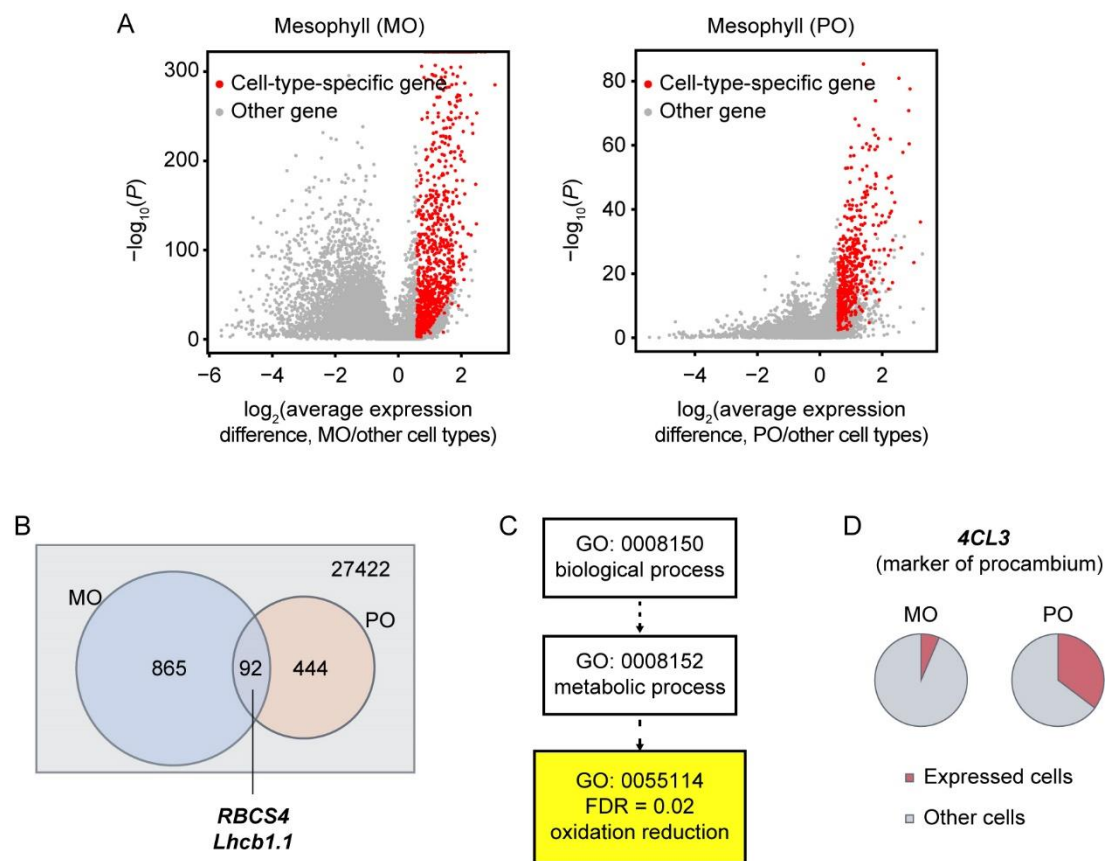
**A:** UMAP visualization of the Leaf samples (cells of two replicates were combined). Dots represent individual cells, colored according to the determined cell type by MICI. The two mesophyll “superclusters” were circled. **B:** Similar to (A), for the Root samples.





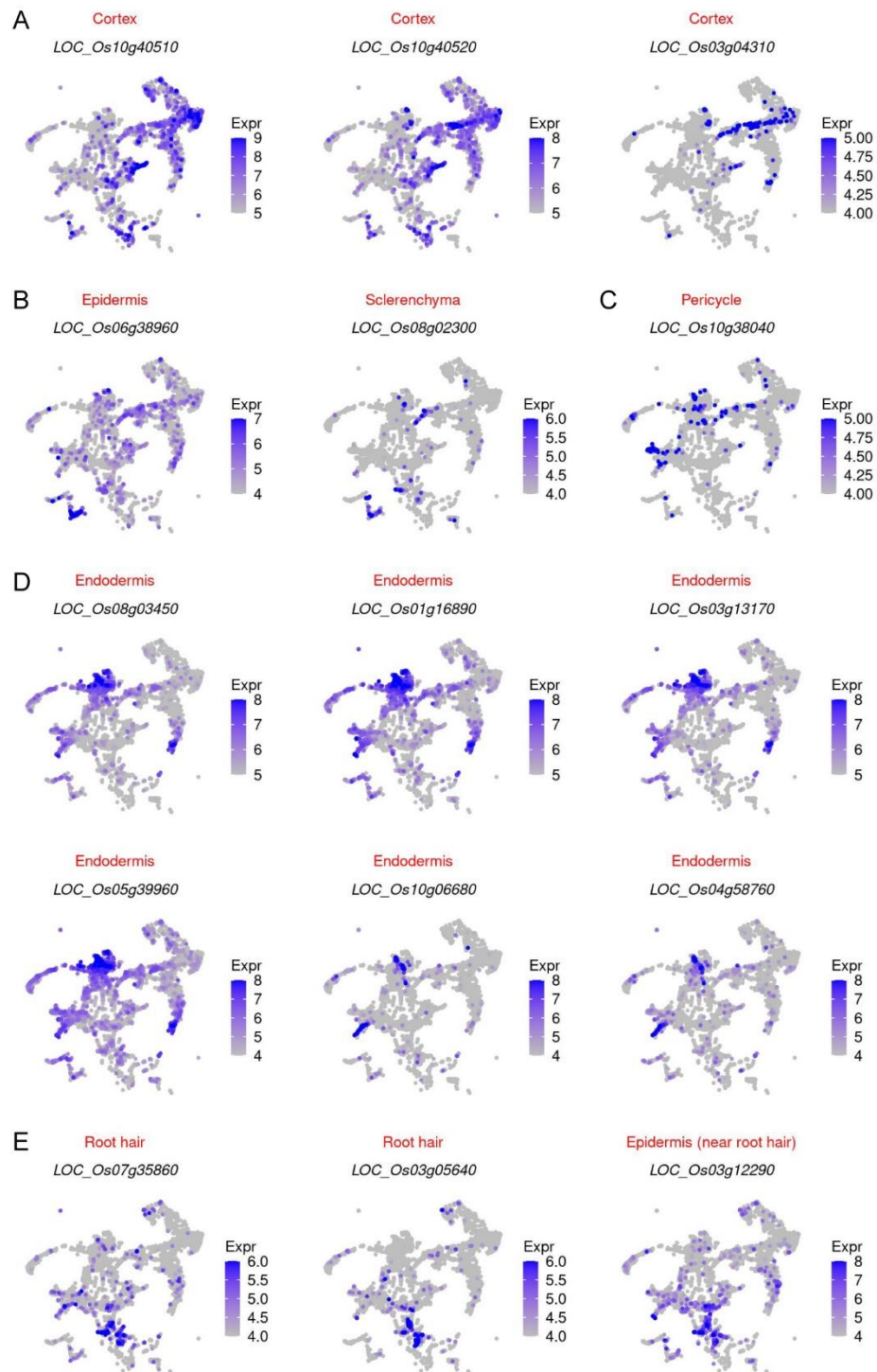
**Fig. S9. RNA *in situ* hybridization assay with the sense probes.**

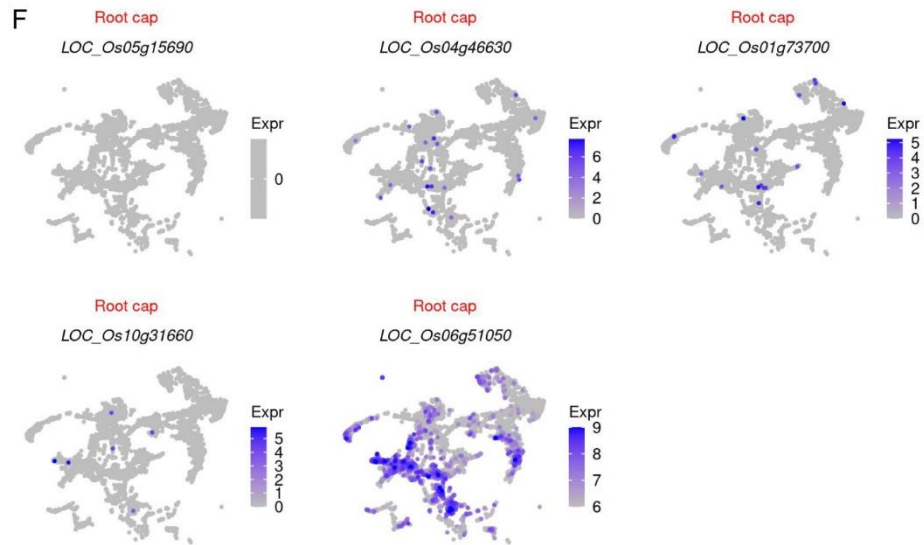




**Fig. S10. The gene expression features of mesophyll cells of two origins.**

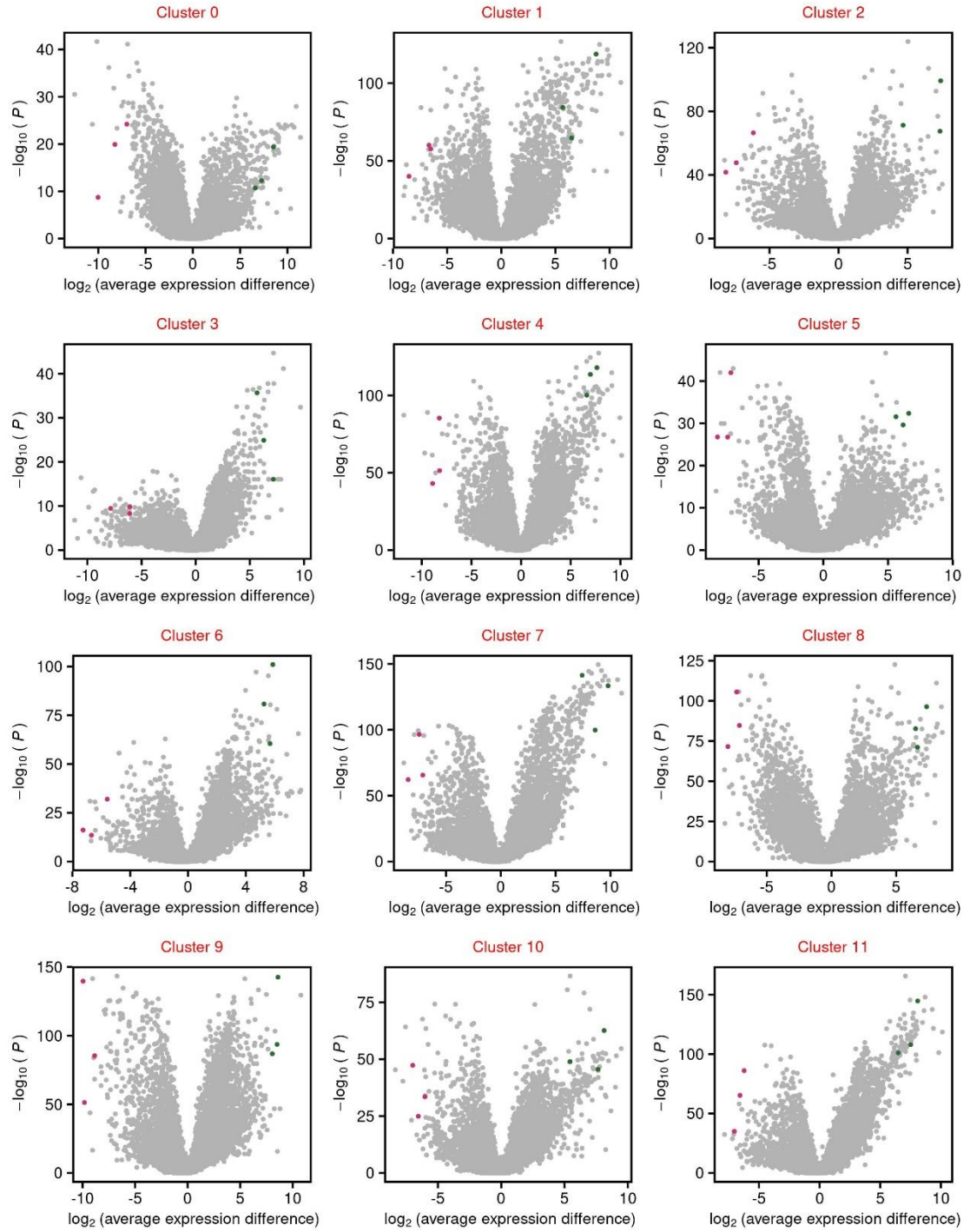
**A:** Scatter plots show the cell-type-specific gene expression in mesophyll cells of mesophyll-precursor origin (MO, showing on the left, including Cluster 21 and 22) and in mesophyll cells of preprocambium origin (PO, showing on the right, including Cluster 4 and 10). Dots represent individual genes, and the identified cell-type-specific genes are colored in red. **B:** Venn plot shows the number of cell-type-specifically expressed genes shared or not shared between mesophyll cells of the two origins. **C:** The enriched GO term (and its parents in biological processes) are shown for the 92 shared cell-type-specific genes between mesophyll cells of the two origins ( $P = 0.0004$ , false discovery rate = 0.02, Fisher's exact test). **D:** The proportion of mesophyll cells that express *4CL3*.

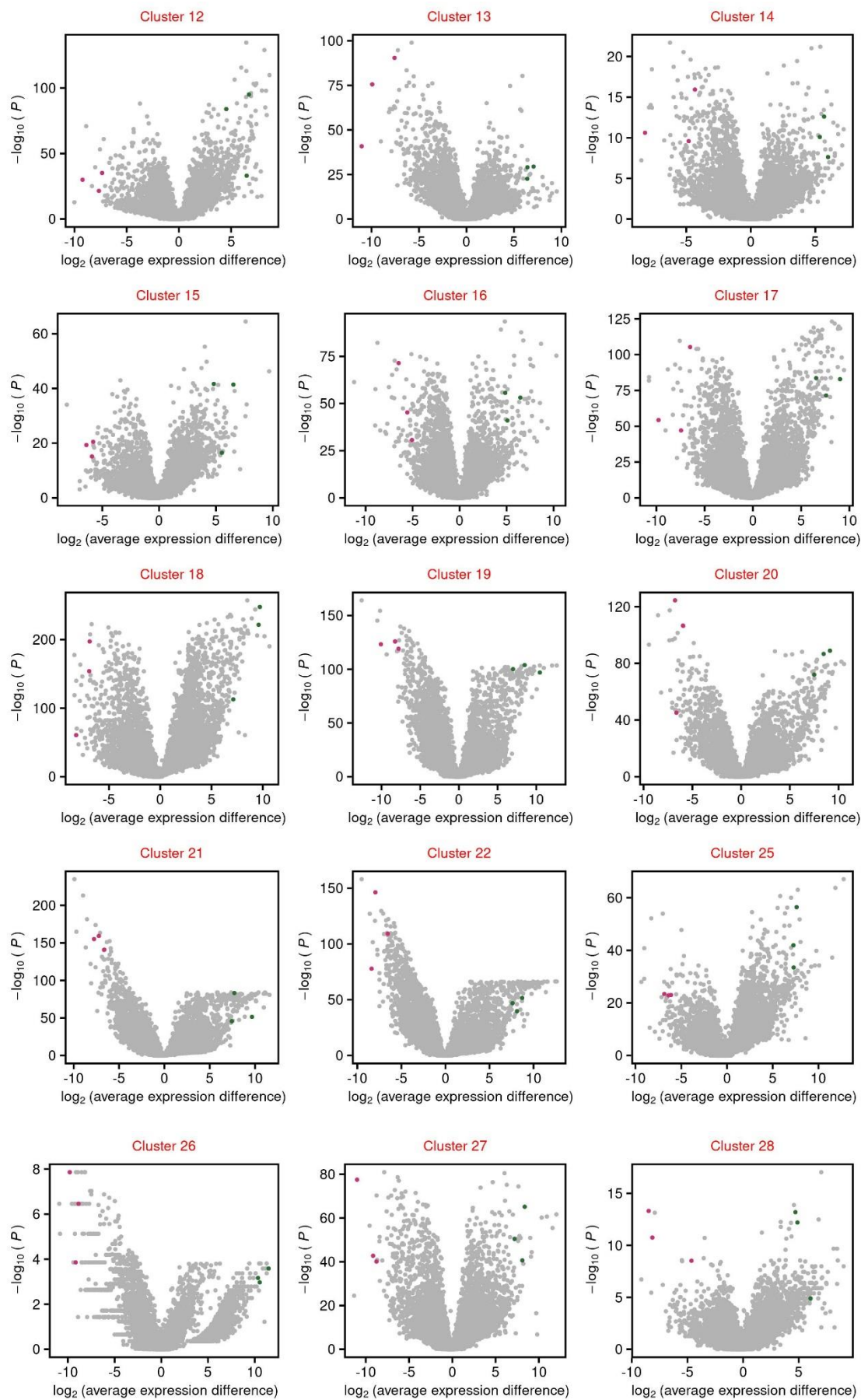




**Fig. S11. UMAP visualization of root marker genes reported by Liu *et al.* (2021) and Zhang *et al.* (2021) for our Root samples (two replicates combined).**

Dots represent individual cells, colored according to the normalized expression level,  $\log(\text{CPM} + 1)$ .

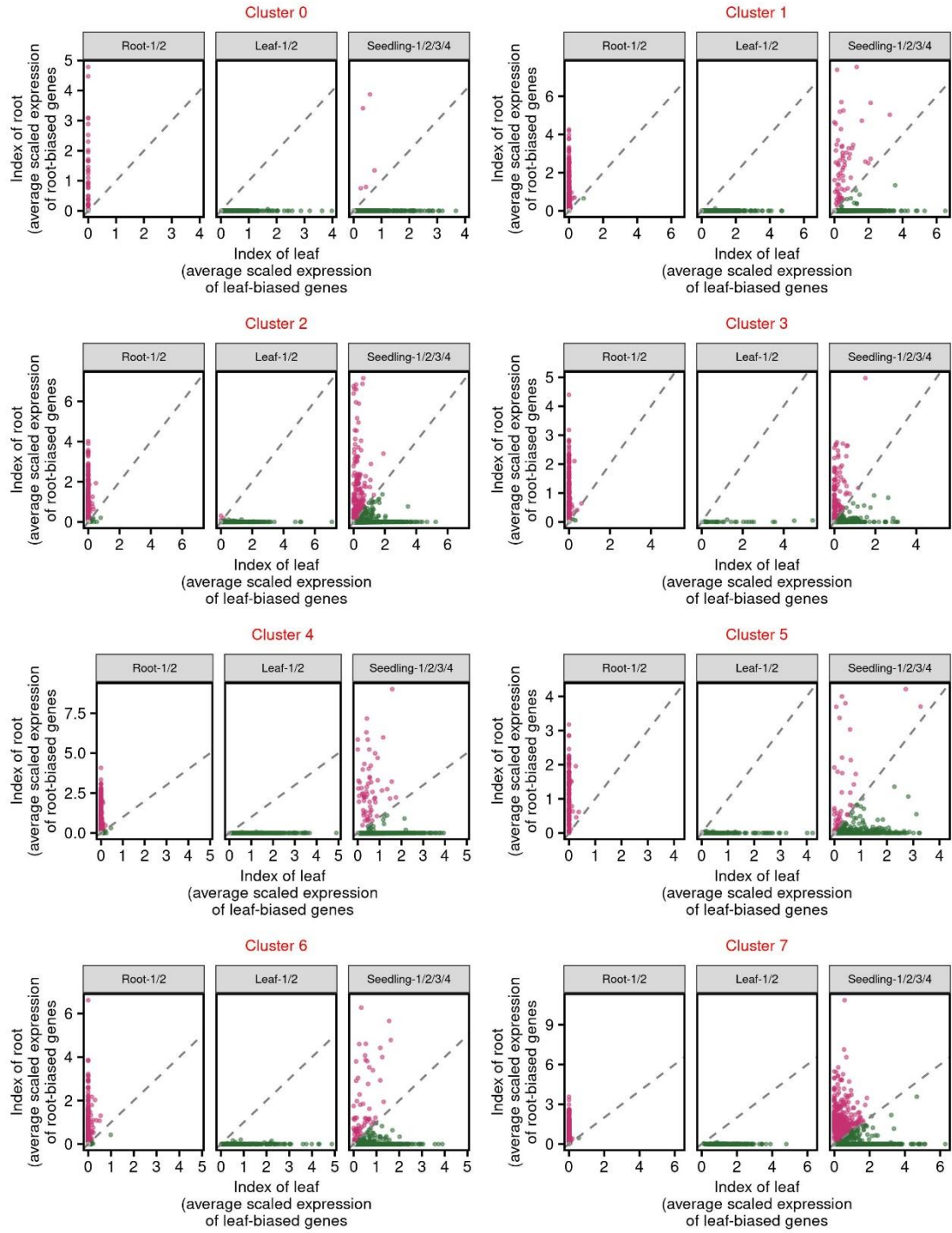


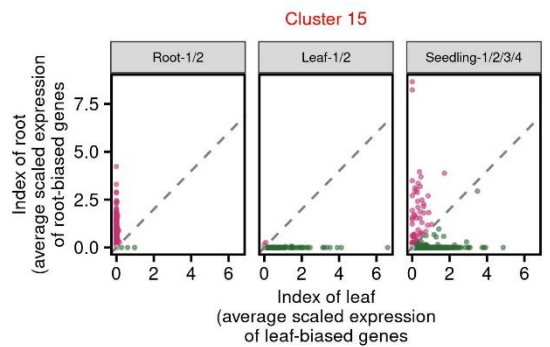
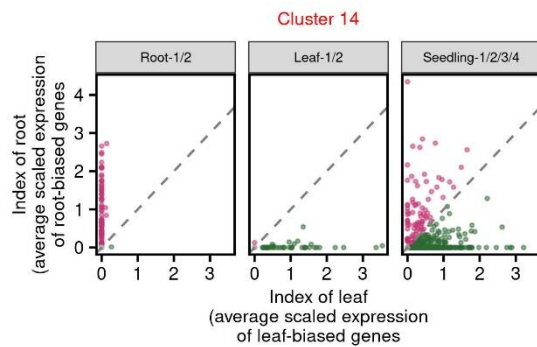
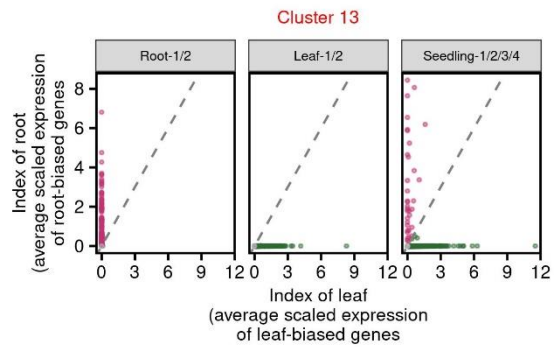
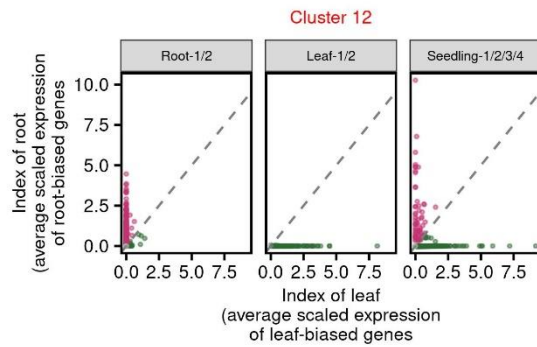
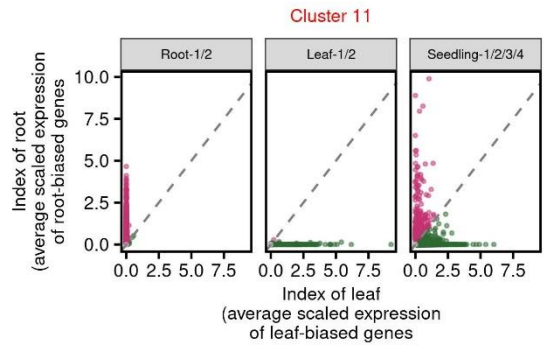
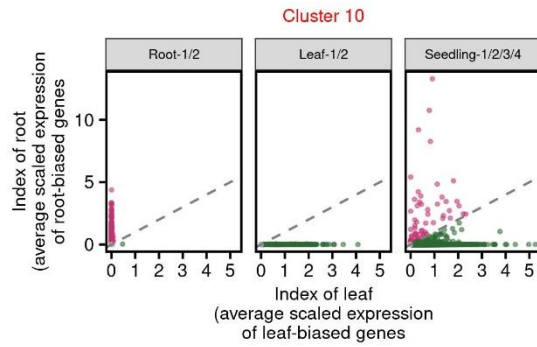
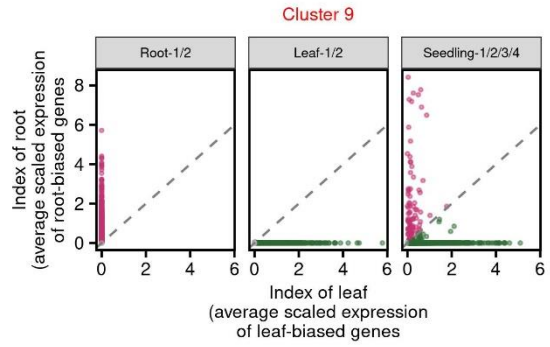
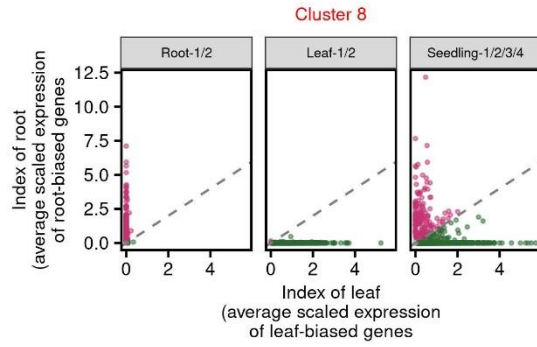


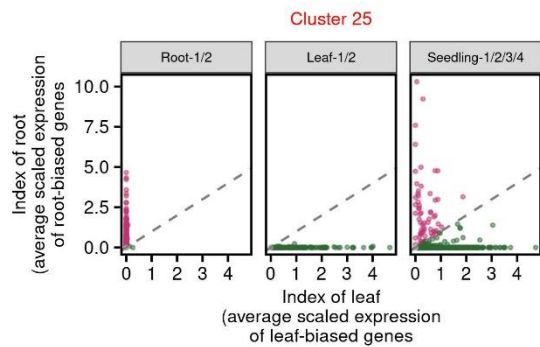
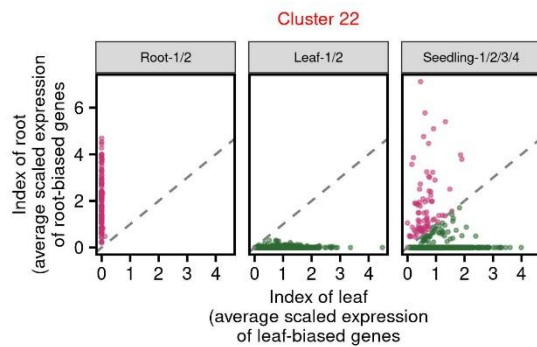
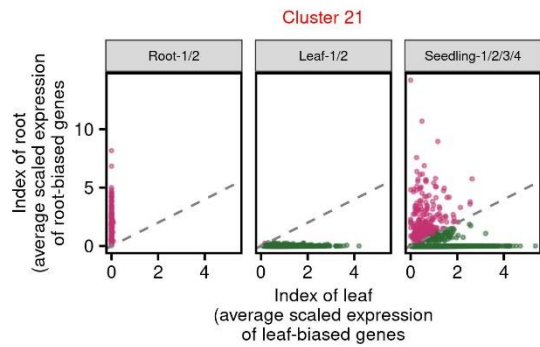
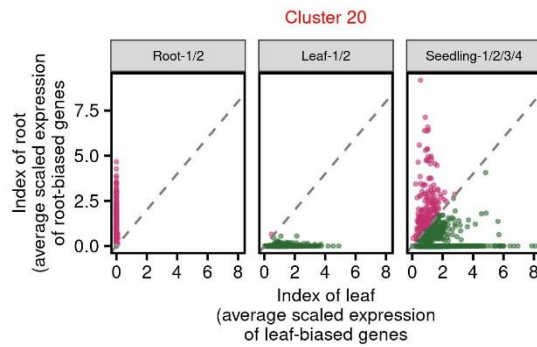
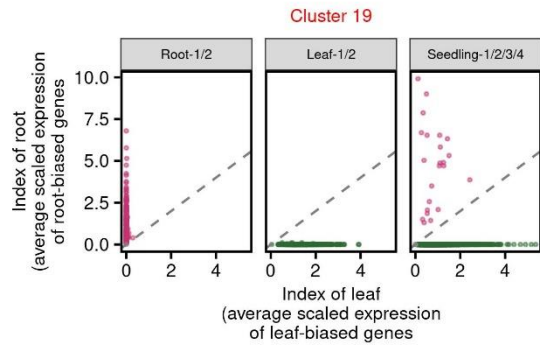
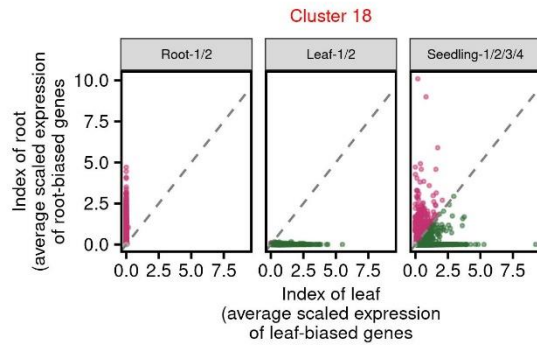
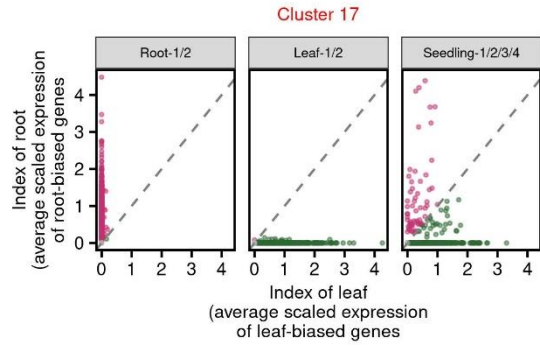
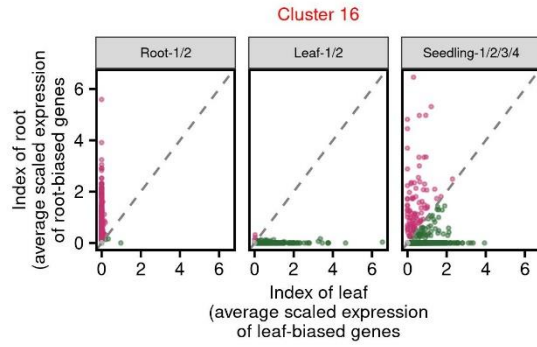
**Fig. S12. Three leaf-biased (green) and three root-biased genes (magenta) in each cluster.**

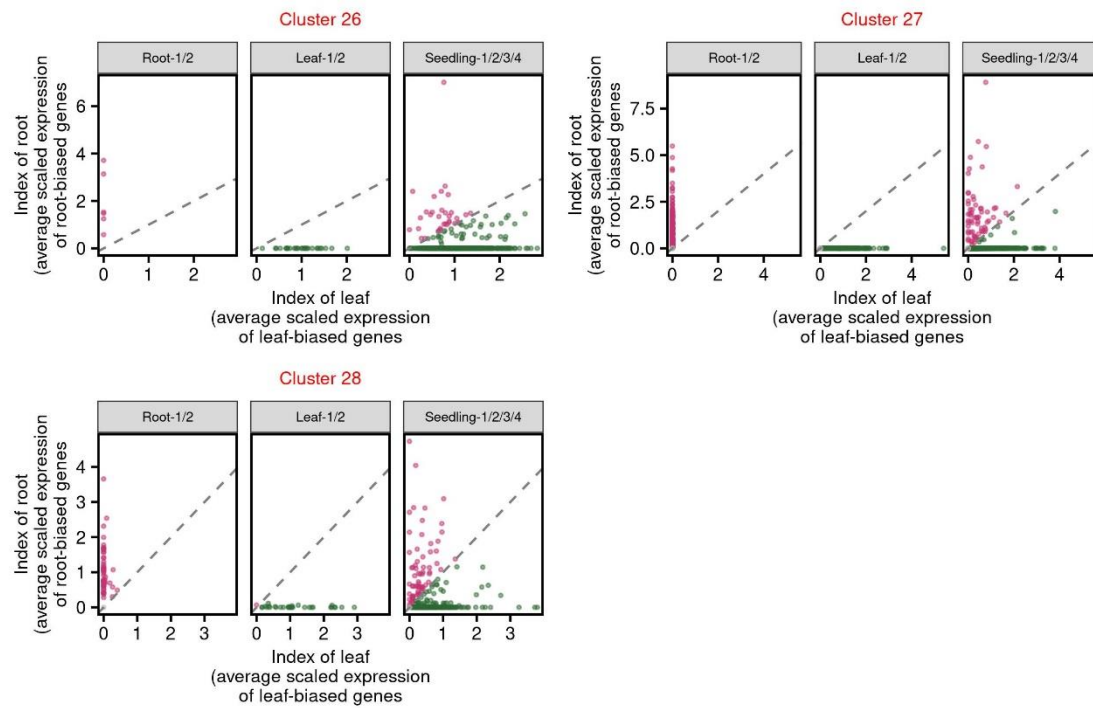
Each dot on the volcano plot represents a gene. The  $\log_2$  transformed ratio on the gene expression level (the average of two replicates, in the unit of CPM) between the Leaf and Root samples is shown on the  $x$ -axis, and the corresponding  $P$  value from the Mann-Whitney  $U$  test is shown on the  $y$ -axis.





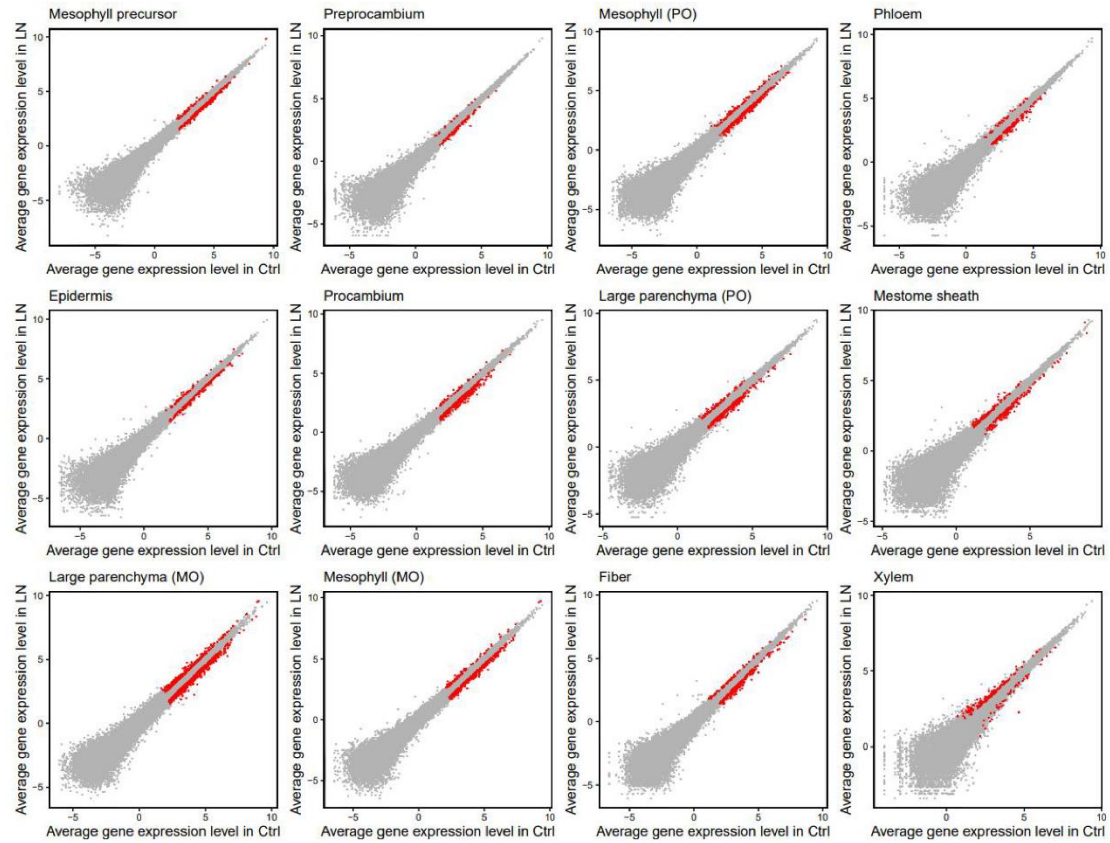




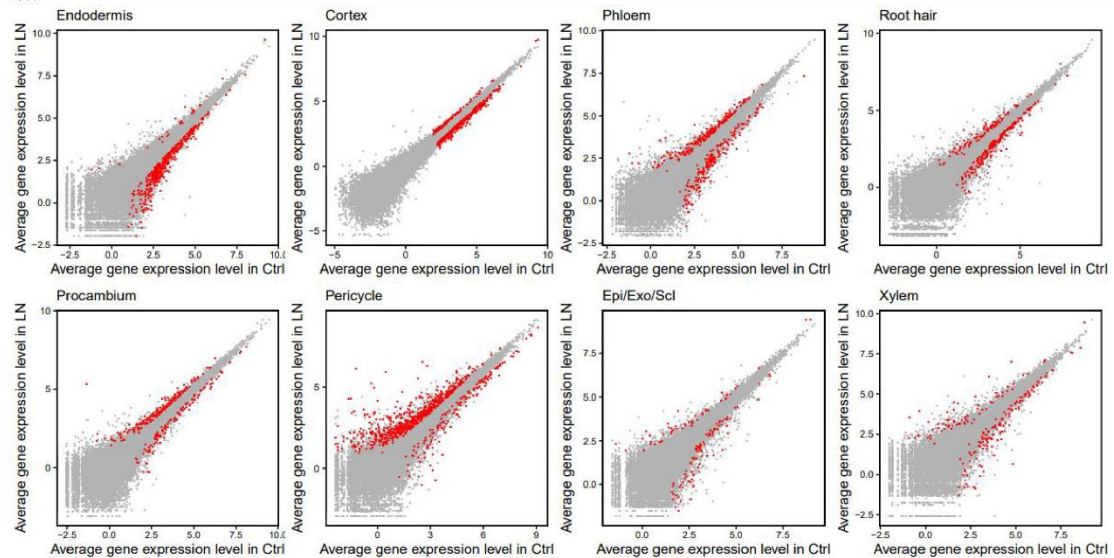


**Fig. S13. The determination of the organ origin (leaf vs. root) for each cell in the Seedling samples.** Similar to Fig. 4D, for all cell clusters.

# Leaf

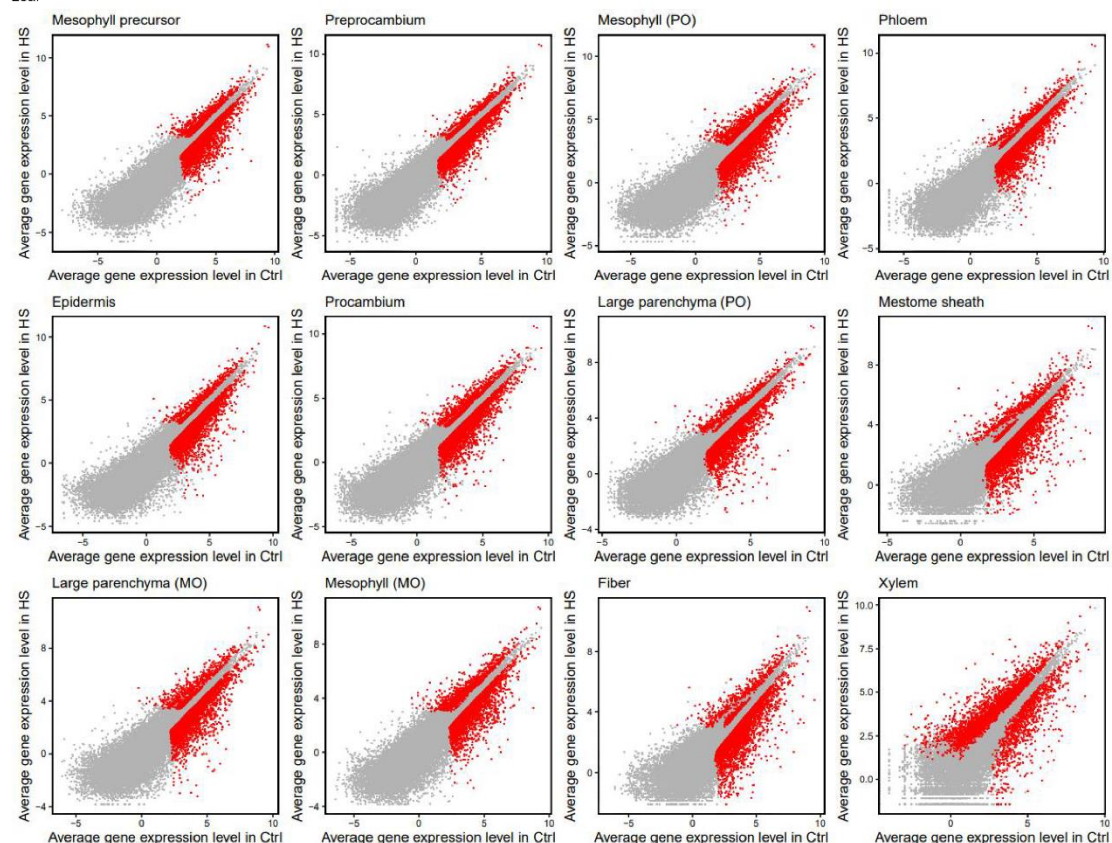


# Root

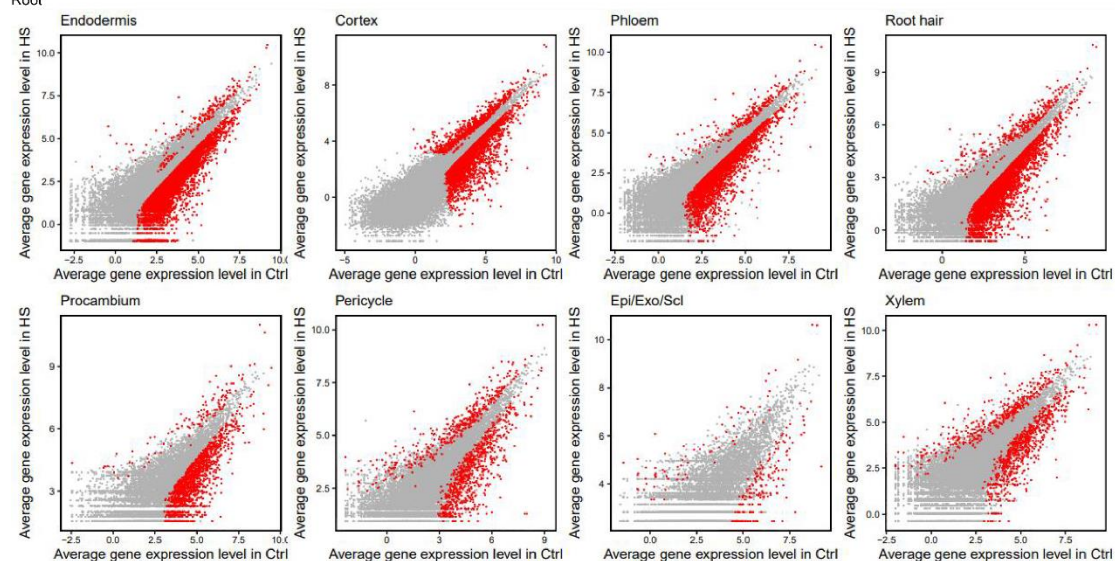




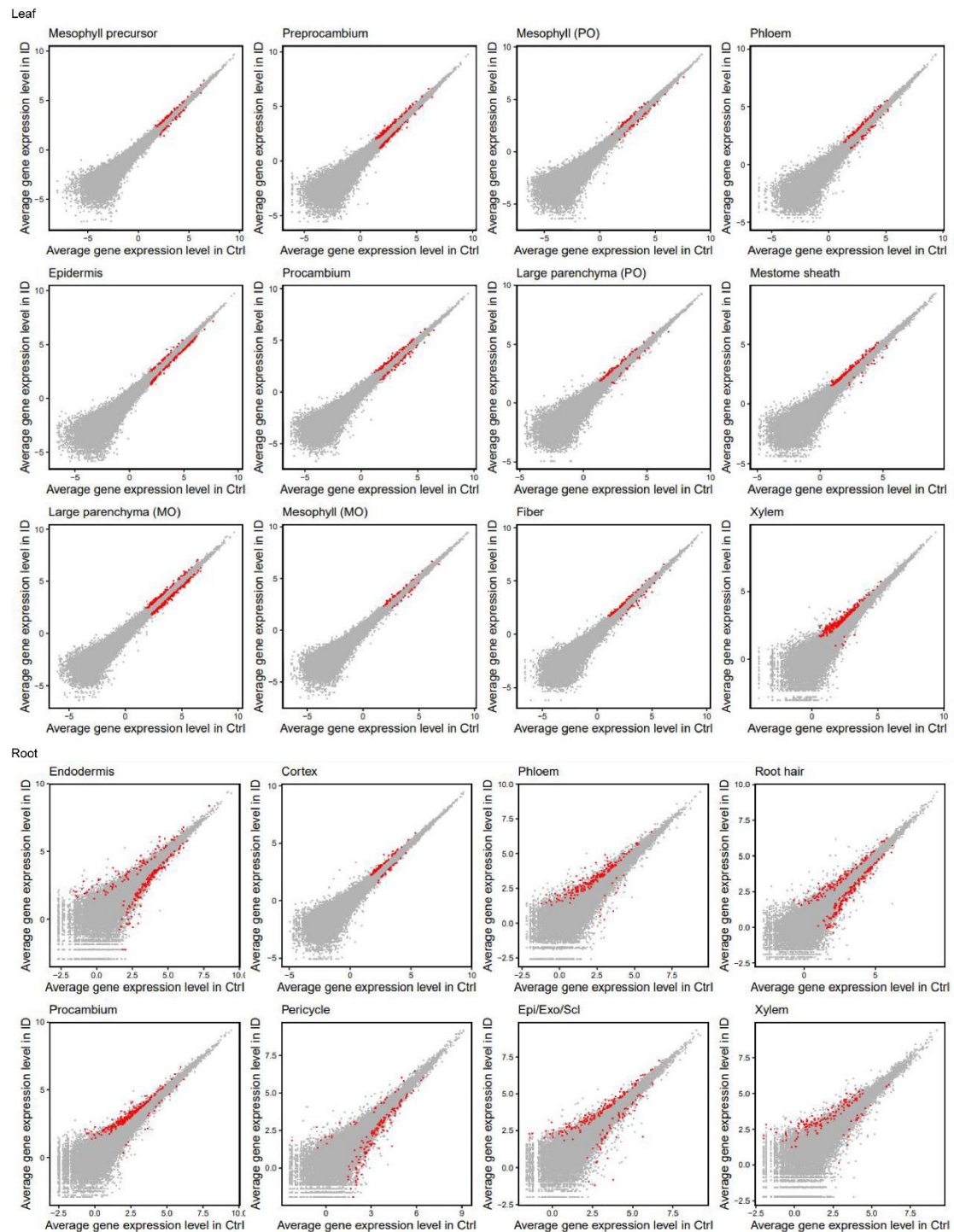
# Leaf



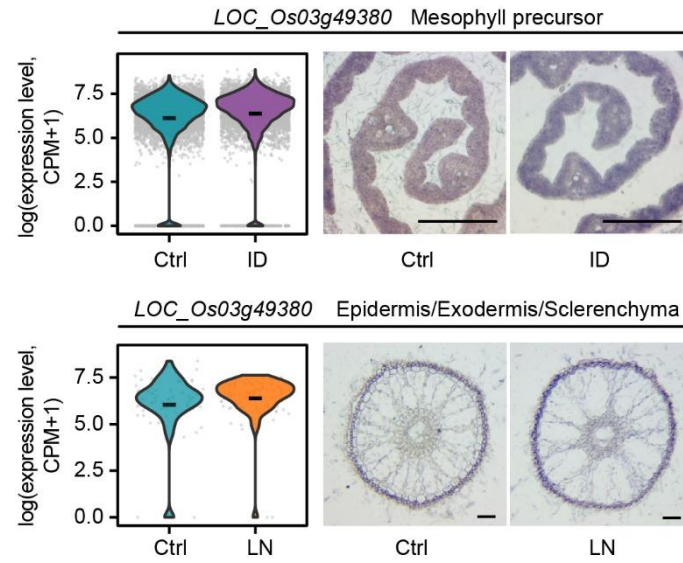
# Root







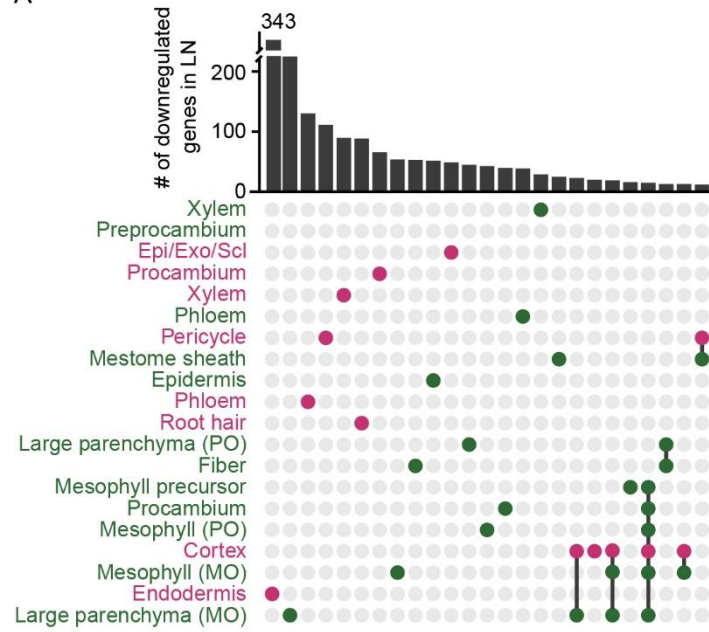
**Fig. S14. Differentially expressed genes identified in each cell type (shown in red dots) upon abiotic stresses.** Similar to Fig. 5D. Ctrl, Kimura B nutrient solution; LN, low nitrogen; HS, high salinity; ID, iron deficiency.



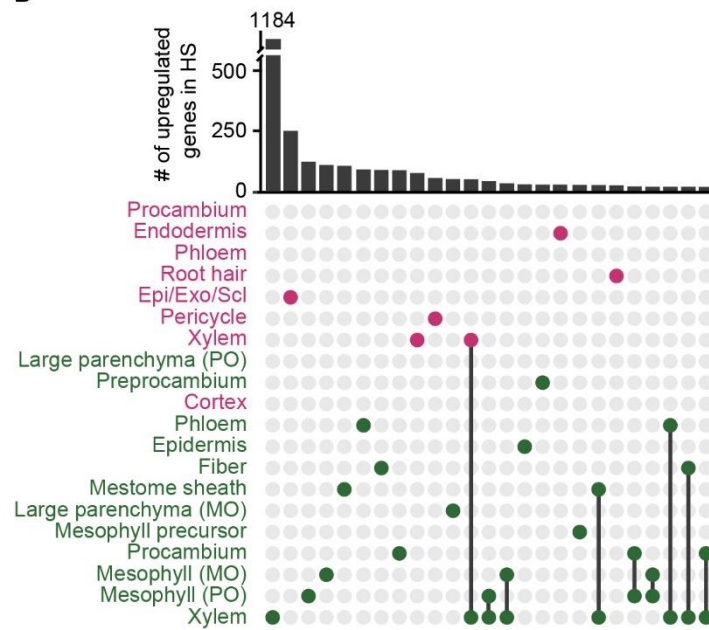
**Fig. S15. RNA *in situ* hybridization assays verified the cell-type-specific differential gene expression upon abiotic stresses.**

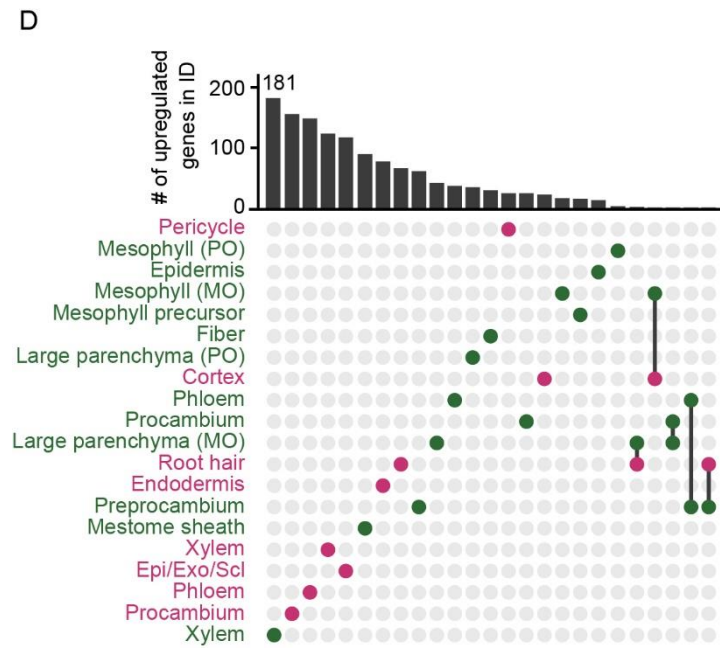
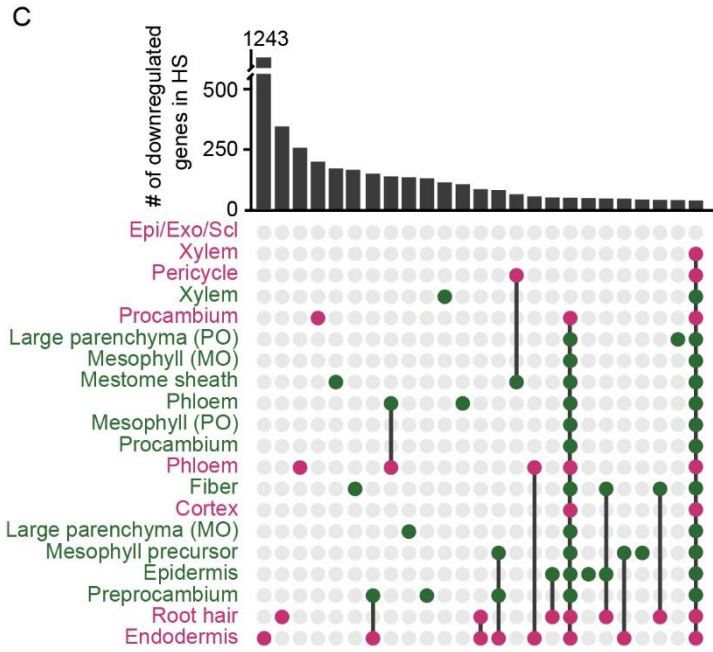
Violin plots show two cases of cell-type-specific differential expression upon abiotic stresses, detected by scRNA-seq (left panel). Each dot represents an individual cell (replicates combined), and the short horizontal line indicates the mean expression level among all cells (top:  $P < 2.2 \times 10^{-16}$ , bottom:  $P = 0.03$ , the one-tailed Mann-Whitney  $U$  tests). Scale bar represents 50  $\mu\text{m}$ .

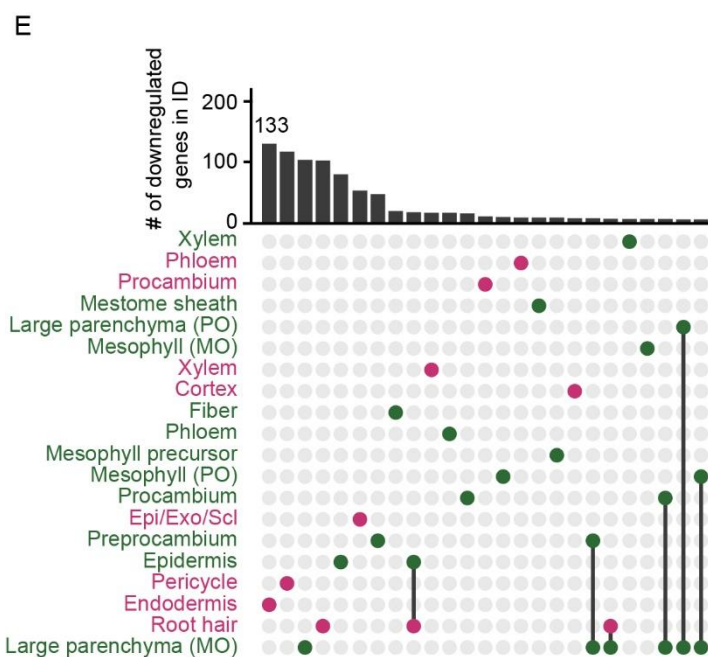
A



B

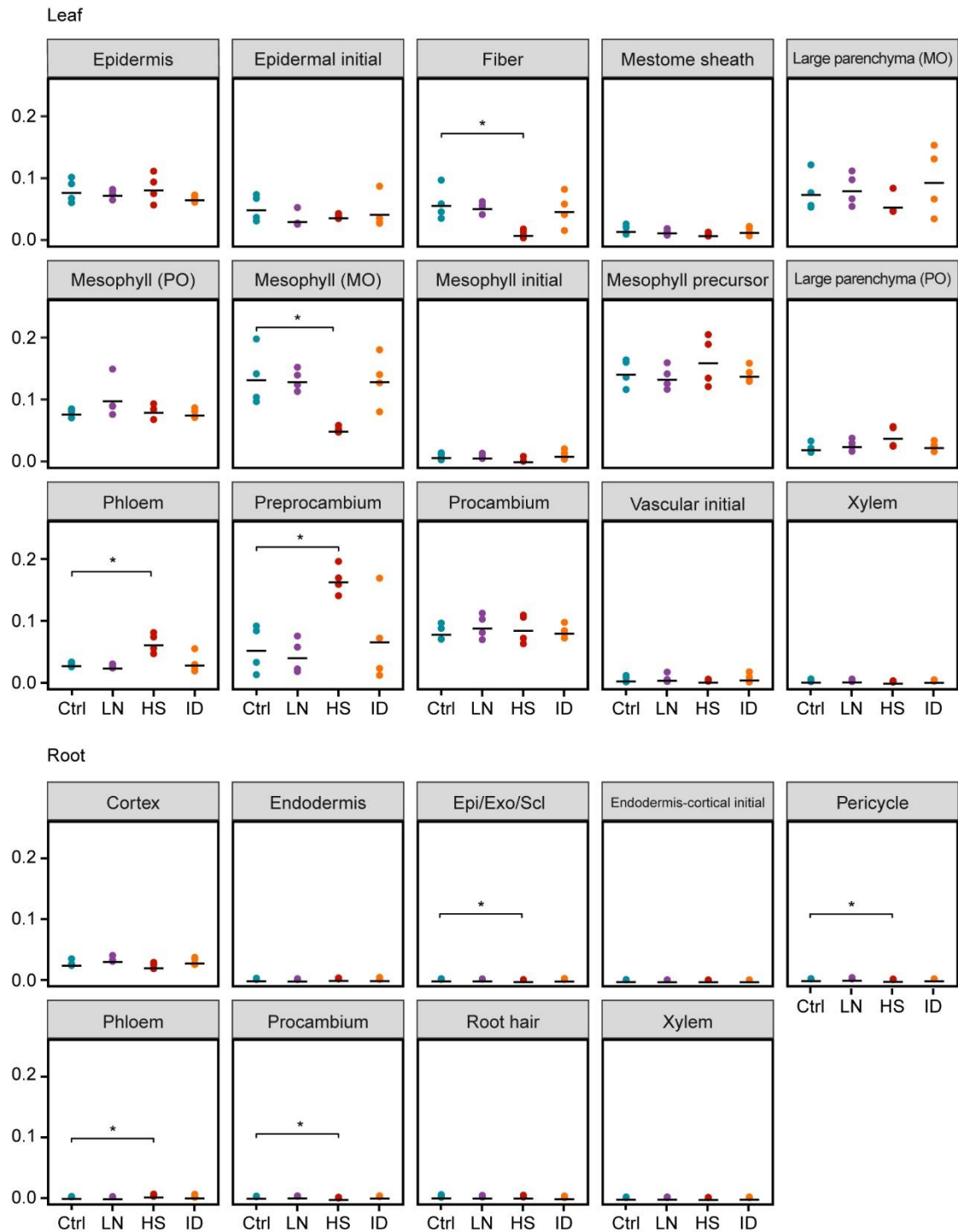




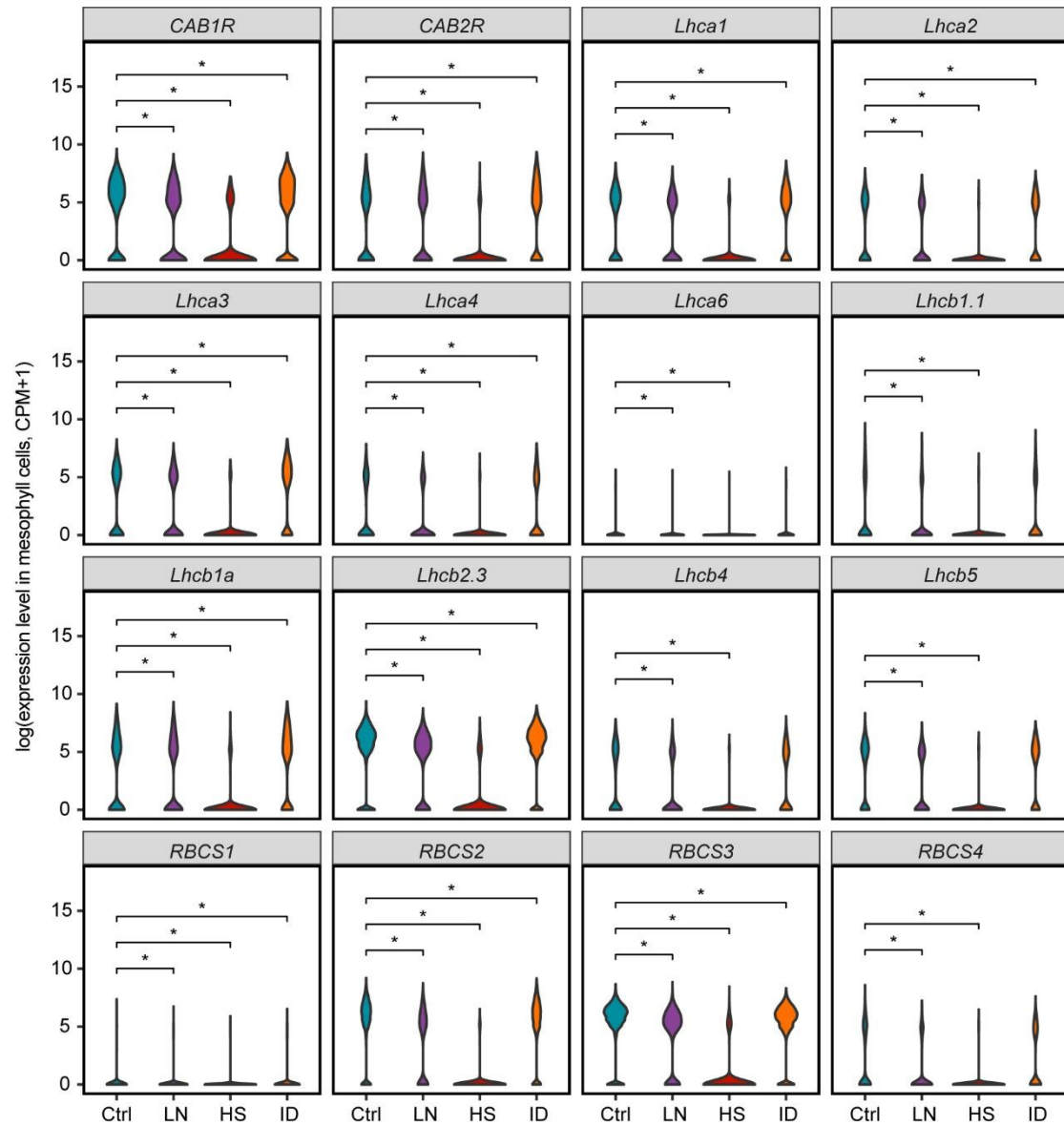


**Fig. S16. The specific or shared transcriptional regulation in response to abiotic stresses among cell types.** Similar to Fig. 5E.

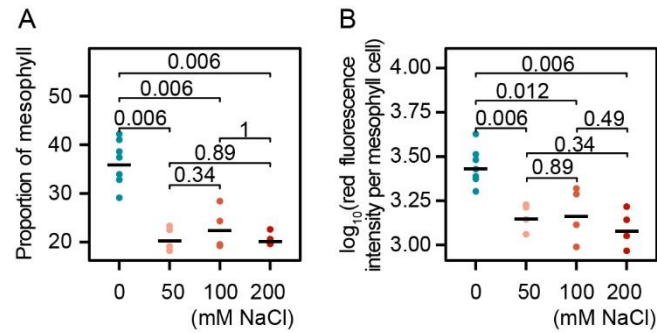




**Fig. S17. Proportional changes in the cell population upon abiotic stresses, detected by scRNA-seq.** Similar to Fig. 6B, including all cell types. \* indicates that  $P$  value (given by the Mann-Whitney  $U$  test)  $< 0.05$ .

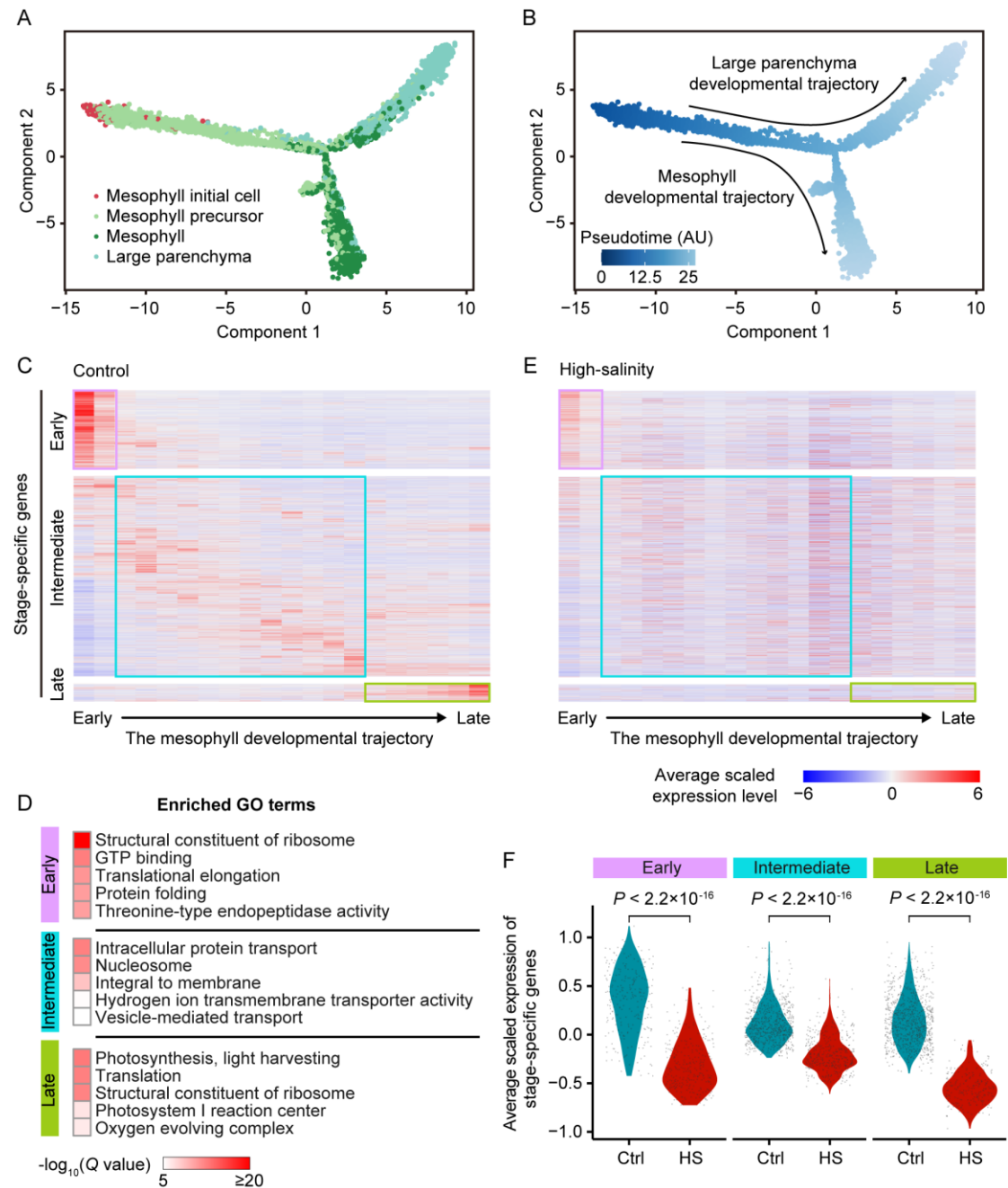


**Fig. S18. The differential expression of genes encoding chlorophyll a/b-binding proteins and the small subunit of RuBisCo upon abiotic stresses. Similar to Fig. 6C. \*** indicates that  $P$  value (given by the Mann-Whitney  $U$  test)  $< 0.05$ .



**Fig. S19. Changes in mesophyll cell population and chlorophyll abundance upon gradient high-salinity stresses.**

**A:** The proportional change in mesophyll cell population upon gradient high-salinity stresses, detected by flow cytometry. [NaCl] was adjusted to the concentration labeled on the bottom, on the basis of the Kimura B nutrient solution (0 mM NaCl). A dot represents the proportion of mesophyll cells among all detected leaf cells for a replicate. The short horizontal line indicates the mean proportion among all replicates of each condition. The numbers of replicates are 7, 4, 4, and 4 for the four conditions. *P* values were given by the Mann-Whitney *U* tests. **B:** Similar to (A), showing the median red fluorescence intensity among mesophyll cells identified by flow cytometry.

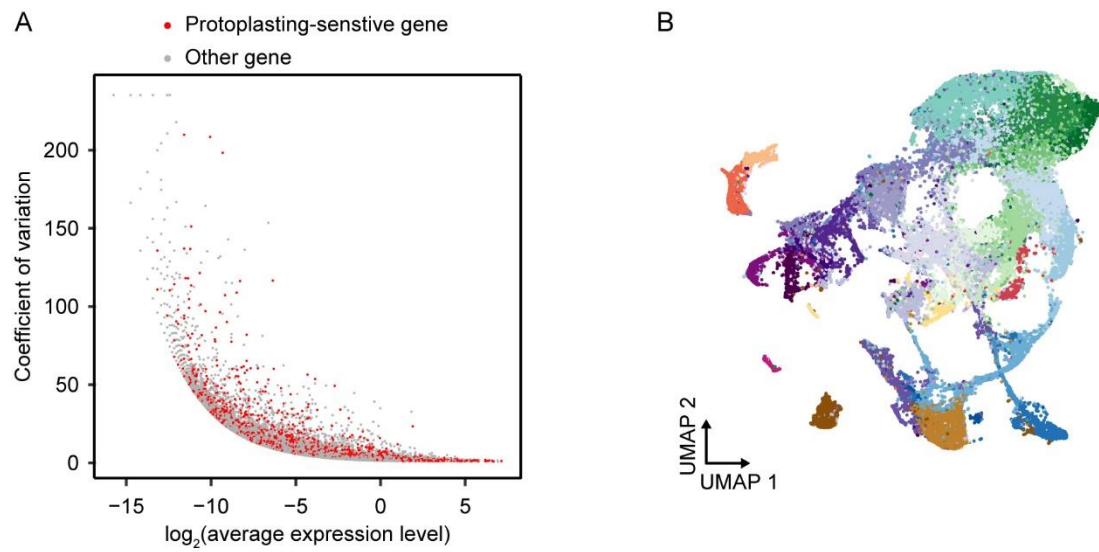


**Fig. S20. The impact of high-salinity treatment on the development of mesophyll cells.**

**A and B:** The development trajectories from mesophyll initial cells to mesophyll or large parenchymal cells. All mesophyll initial cells, mesophyll precursors, mesophylls, large parenchyma were retrieved from the 16 Seedling samples of the four conditions. Each dot represents a cell and is colored according to its cell type (A) or the pseudotime estimated by Monocle (B). **C:** The heat map shows the scaled expression levels of the 7766 highly dynamic genes over the mesophyll developmental trajectory in the four

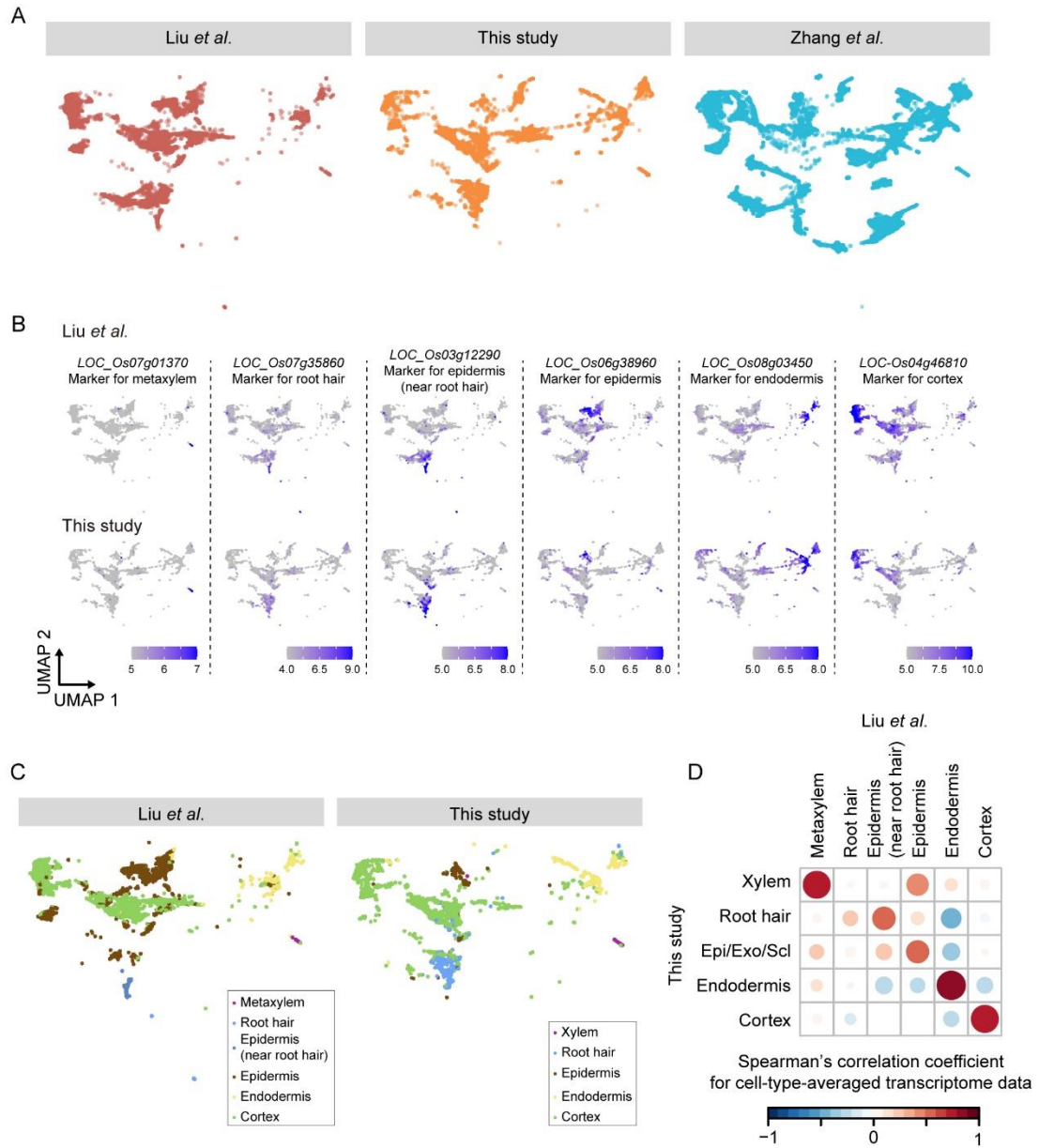
control samples. Each line represents a gene, and each column represents one of the 20 equal-distance cell bins according to the pseudotime, ordered from early to late in development. The color in the heat map shows the average scaled expression level among all cells in a bin. Genes were grouped into three classes (early, intermediate, and late) by the function “hclust” in *R*, based on their temporal expression patterns. The early, intermediate, and late cells in the mesophyll developmental trajectory were defined based on the specific expression of these three classes of genes (shown in boxes). **D**: Top five enriched GO terms are shown for each gene class ( $P < 7.9 \times 10^{-7}$ , false discovery rate  $Q < 3.7 \times 10^{-5}$  for all terms, Fisher’s exact test). **E**: Similar to (C), for cells in the four high-salinity samples. The same genes and the same cell-bin ranges are presented as in (C). **F**: The violin plots show the average scaled expression levels of early, intermediate, and late genes (shown in individual dots) in the respective cells along the mesophyll developmental trajectory. *P* values are given by the Mann-Whitney *U* tests.





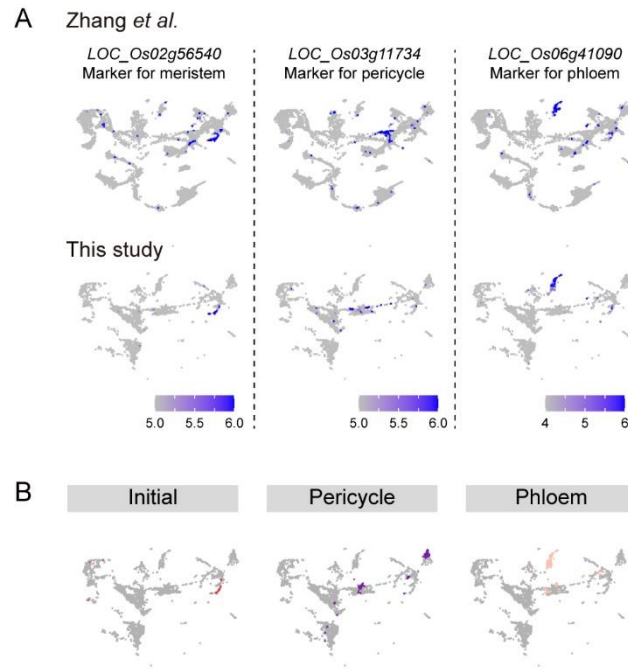
**Fig. S21. The influence of protoplast isolation on data analysis.**

**A:** The Scatter plot shows the coefficient of variation for the expression level (UMI count) among cells for each gene, against its average expression level. **B:** Similar to Fig. 2B, visualization of the single-cell transcriptomes of rice seedlings (Leaf-1/2, Root-1/2, and Seedling-1/2/3/4) on the UMAP graph, without protoplasting-sensitive genes being removed. Cells were colored according to Fig. 2B.



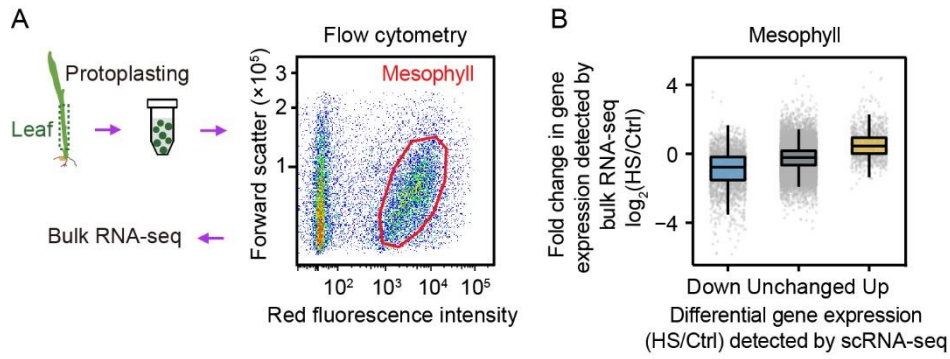
**Fig. S22. The comparison of the cell-type annotations among three studies.**

**A:** The UMAP graphs show the results of single-cell transcriptome data normalization, integration, and dimensionality reduction for rice root cells among our study (Root-1/2 combined), Liu et al. (2021, Nipponbare), and Zhang et al. (2021a, ZH11). **B:** The expression patterns of six cell-type-specific genes reported in Liu et al. (2021). **C:** The direct comparison of the cell annotation between our study and Liu et al. (2021) on the UMAP graphs. **D:** The pairwise Spearman's correlation coefficients for the expression profile between the cell types identified in our study and Liu et al. (2021). The figure is otherwise similar to Fig. 4A.



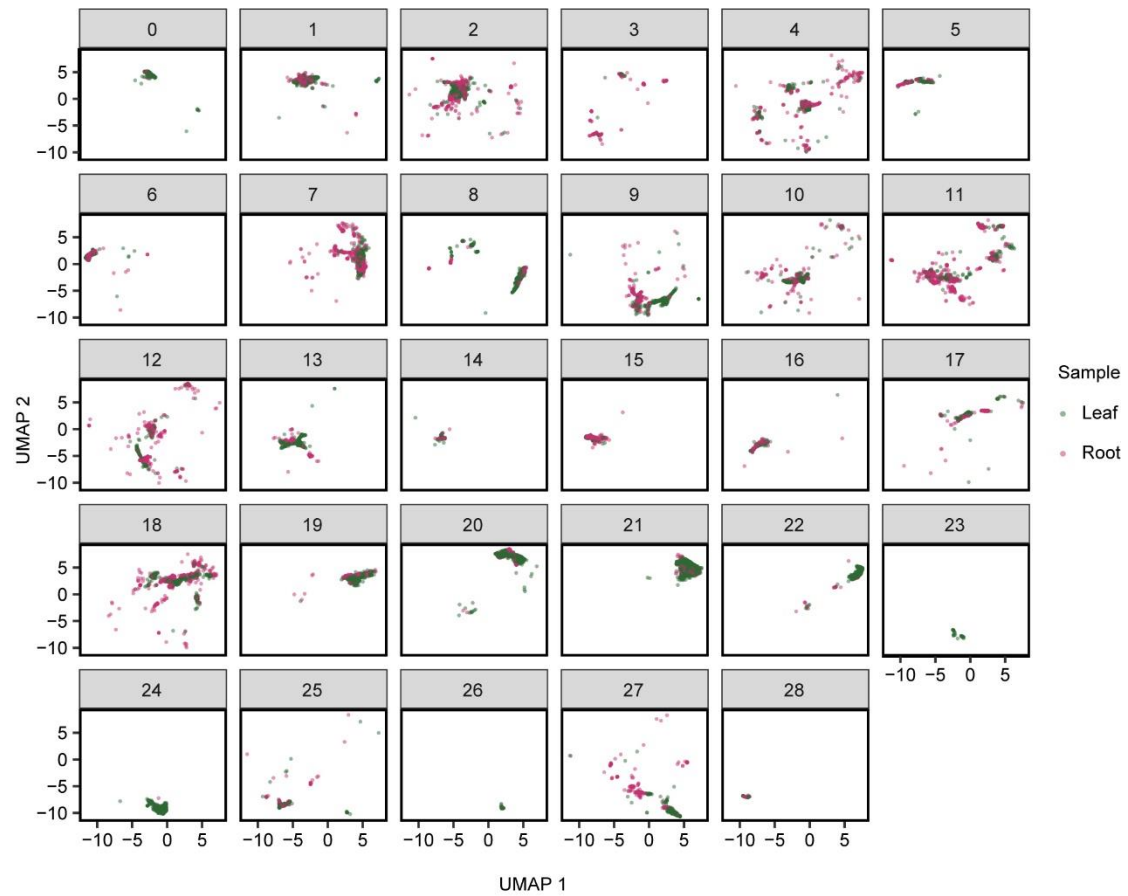
**Fig. S23. The comparison of the root cell annotation between our study and Zhang et al. (2021a).**

**A:** The expression patterns of three cell-type-specific genes reported by Zhang et al.. These three cell types were not reported by Liu et al. **B:** Individual cells of the three corresponding cell types annotated in our study were labeled on the UMAP graph.



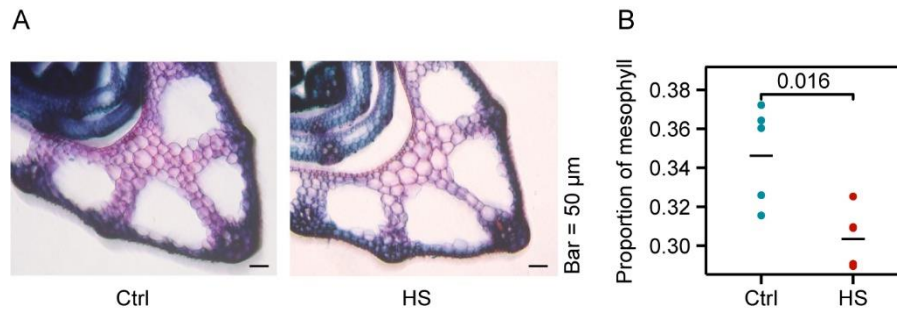
**Fig. S24. Differential gene expression in response to the high-salinity treatment in mesophyll cells, detected by bulk RNA-seq on the sorted cells or by scRNA-seq.**

**A:** A schematic shows the experimental workflow of the bulk RNA-seq on the sorted mesophyll cells. Protoplasts were isolated from leaves of two-week-old rice seedlings and loaded onto the flow cytometer. Mesophyll cells were sorted according to both forward scatter and red fluorescence intensity (gate shown in red). The sorted cells were used for the extraction of total RNA and the construction of bulk RNA-seq libraries. **B:** Differential gene expression in mesophyll cells detected by scRNA-seq was verified by the bulk RNA-seq on the sorted cells. The upregulated (“Up”) and downregulated (“Down”) genes were defined in Fig. S14. The expression ratio of each gene, estimated by bulk RNA-seq, is shown on the y-axis ( $P < 2.2 \times 10^{-16}$  in all pairwise comparisons, the Mann-Whitney  $U$  tests).



**Fig. S25. UMAP visualization of cells in individual clusters in the Leaf and Root samples.**

Cells from individual clusters were separately drawn in the same UMAP space. Dots represent individual cells, colored according to the materials (Leaf or Root).



**Fig. S26. The proportional change in the mesophyll cell population, upon the high-salinity stress, detected by the fast green/safranin O staining.**

**A:** Leaf cross-sections showing the morphological features of rice seedlings grown in the Kimura B solution (Ctrl, 0 mM NaCl) and under the high-salinity treatment (HS, 200 mM NaCl). Mesophyll and large parenchymal cells were stained blue and red by fast green/safranin O, respectively. **B:** The proportional changes in mesophyll cell population upon high-salinity stress, detected by histology. Dots represent replicates, and the short horizontal line indicates the mean among four replicates of each condition.